



Same, Same but Different? A Multi-Method Review of the Processes Underlying Executive Control

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Abstract

Attention, working memory, and executive control are commonly considered distinct cognitive functions with important reciprocal interactions. Yet, longstanding evidence from lesion studies has demonstrated both overlap and dissociation in their behavioural expression and anatomical underpinnings, suggesting that a lower dimensional framework could be employed to further identify processes supporting goal-directed behaviour. Here, we describe the anatomical and functional correspondence between attention, working memory, and executive control by providing an overview of cognitive models, as well as recent data from lesion studies, invasive and non-invasive multimodal neuroimaging and brain stimulation. We emphasize the benefits of considering converging evidence from multiple methodologies centred on the identification of brain mechanisms supporting goal-driven behaviour. We propose that expanding on this approach should enable the construction of a comprehensive anatomo-functional framework with testable new hypotheses, and aid clinical neuroscience to intervene on impairments of executive functions.

Keywords Attention · Executive control · Working memory · Dysexecutive syndrome · Clinical anatomical correlations

Introduction

Attention, working memory, and executive control are facets of human cognition that are typically considered to be distinct but related (Stuss & Alexander, 2007). In this review article we attempt to analyze this claim, and we begin by describing how these functions are conceptualized in hallmark neurocognitive models.

A closer look to the most common ways of defining these three psychological constructs raises questions of whether (or more likely, to what extent) these constructs are one and the same. We then integrate human lesion studies with recent behavioural, invasive, and non-invasive multimodal neuroimaging and stimulation evidence to provide an updated understanding of the relationship between these processes in the human brain.

Attention refers generally to the selection and maintenance of focus on one of many possible internal or external stimuli (James,

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1907). The central role of attention in everyday cognition is captured in a plethora of attentional models (Buschman & Kastner, 2015; Petersen & Posner, 2012), and is supported by evidence showing that abnormal functioning of attention is associated with a variety of neurological (Bartolomeo et al., 2012) and psychiatric conditions (Cornblatt & Keilp, 1994; Mackie & Fan, 2017; Paelecke-Habermann et al., 2005; Spagna et al., 2018, 2022; Stuss et al., 1981).

One of the earliest models of attention, the *attentional control of action* model of Norman and Shallice (1980), proposed the existence of two attentional control processes: (a) the contention scheduling mechanism that selects between conflicting routine actions, and (b) the supervisory attentional system (SAS) that intervenes when the selection of routine actions is not sufficient (as in novel situations). This model draws a direct link between attention and cognitive control, with SAS deficits being responsible for the generation of a dysexecutive syndrome.

Another influential theory of attentional functions described three separate but interacting components of attention (Posner & Petersen, 1990) including: (a) alerting, or maintaining of an alert state and readiness to react, that can be *phasic*^G (see definition in Glossary; glossary definitions are henceforth indicated by a superscript “^G”) or *tonic*^G, (b) orienting, or the selection of stimuli, that can be *exogenous*^G (i.e., automatically driven towards a salient stimulus) or *endogenous*^G (i.e., driven by voluntary processes implemented in a top-down fashion), and (c) executive control, which plays a role in goal-directed behaviour, error detection, conflict resolution, and inhibition of automatic responses. The executive control network was initially characterized in terms of response selection, but later was presented as *executive attention* (Fernandez-Duque & Posner, 2001; Posner & DiGirolamo, 1998) and described as the intentional or voluntary control over behaviour. Complementary recent work demonstrated that executive control is a supramodal (Spagna et al., 2015, 2018, 2020) and supra-domain function of attention (Wu et al., 2020), supporting the inhibition of competing messages from irrelevant sources and the selection of relevant inputs from relevant sources, irrespective of the sensory modality. Along similar lines, Duncan (2013) proposed the existence of a domain-general multiple demand (MD) system responsible for the management of different “cognitive steps”. In this model, *attentional episodes*^G are at the core of the organisation of complex behaviour, and the MD system is recruited to identify the correct task set, to order the sequence, and shifts between task sets^G, all of which are functions lying between executive control and working memory (Duncan, 2013).

More recent models of attention shifted their focus towards portraying attention as a periodically changing process and suggest that this periodicity determines shifts in attentional state between focus and distraction or reorienting (Buschman & Kastner, 2015; Fiebelkorn & Kastner, 2019; Helfrich & Knight, 2016; Wilson et al., 2010). Together, these models demonstrate that competition

for attentional resources occurs in many dimensions and attentional control processes are necessary to ensure that the relevant stimuli and processes gain priority. Other models have been developed to encompass additional aspects pertaining human attentional abilities, such as whether attention is oriented inward towards internally generated information (e.g., memory content) or outward toward sensory information coming from the environment (Chun et al., 2011), or the motivational dimension that can affect the distribution of attentional resources (e.g., Kurzban et al., 2013). Focusing on the external–internal dichotomy of voluntary attention, Burgess et al. (2007) suggested that a “supervisory attentional gateway” would enable individuals to control their engagement with external (i.e., environmental stimuli) or internal (i.e., self-generated or maintained representations) information. Adding this dimension to attentional models led to the integration of *mind-wandering* states^G to which attentional resources can be allocated (Chun et al., 2011). Along these lines, the resource-control theory (Thomson et al., 2015) proposes the existence of a continuous bias of attentional resources toward mind wandering^G type states, with a competitive relationship between task-related attention and mind-wandering^G. From this perspective, executive control of attention toward external or environmental goals fails when attentional resources are devoted instead toward internally generated or mind-wandering states.

Working memory involves temporary maintenance and manipulation of information in the mind in a goal-driven context (Baddeley, 1992). The most prominent cognitive model of working memory (Baddeley, 2012) identifies four components: (a) the *central executive*, a limited-capacity processing system, (b) the *phonological loop*, a processing and storage system for phonological information, (c) the *visuo-spatial sketchpad*, a processing and storage system for visuo-spatial information and (d) the *episodic buffer*, a storage system holding information from phonological loop, visuo-spatial sketchpad and the long-term memory system. These components allow for distinguishing between a verbal working memory system, which is important for temporarily storing and manipulating verbalizable information, and a spatial working memory system, which is important for temporarily storing and manipulating visuo-spatial information. The link between attention, working memory, and another construct called cognitive control, is explicit in this model because the central executive is considered to act as an attentional controller, hence being the most important component of the working memory system, albeit the least understood (Logie, 2016). The central executive is involved in the manipulation and control of the information held in working memory and is critical for a variety of cognitive operations such as coordination, inhibition, preventing attentional capture by irrelevant stimuli, switching, and updating (Baddeley, 1996; Miyake et al., 2000; Friedman & Miyake,

2004), exemplifying how cognitive control and working memory are intertwined. Further, neuroanatomical and functional studies seem to indicate that the central executive component of Baddeley and Hitch (1974) includes a class of top-down functions that exert control over more elementary processes, named *executive functions*. Of note, the central executive component of Baddeley and Hitch (1974) model is not equivalent to the executive functions, with the latter term proposed by Lezak (1982) on the basis of more specific processes identified by Luria (1966) (i.e., goal-setting, action initiation, planning, execution, and verification).

In their model, Miyake et al., (2000) propose the existence of three correlated but distinct functions: (a) shifting between mental sets or tasks, (b) updating and monitoring of working memory contents, and (c) inhibiting irrelevant dominant responses. These functions are central, supramodal functions that operate in non-routine situations (e.g. during novel, conflicting, or complex tasks) or in routine situations when self-activation is required to optimize efficiency (Godefroy et al., 2018). Other studies showed that the attentional controller of the working memory system is involved in both “cold” processing and “hot” affective processing (Hofmann, Schmeichel, & Baddeley, 2012; Ochsner & Gross, 2005).

More recently, a class of cognitive models of working memory termed “state-based models” was put forth, positing that information maintained in working memory exists in one of several states of activation established by the allocation of attention (D’Esposito & Postle, 2015). Different state-dependent models of working memory vary in the type of information held in working memory such as sensory-motor information (e.g., Awh & Jonides, 2001) or semantic and symbolic information (e.g., Cowan, 1998; Oberauer, 2002, 2009). Yet, all these models state that attentional selection of mental representations is the process responsible for bringing the information into working memory and thus explains working memory properties, including capacity limitations and proactive interference from no-longer-relevant items (D’Esposito & Postle, 2015).

The term *executive control*, often used interchangeably with *cognitive control* and *attentional control* and at times conflated with the term *executive functions*, refers broadly to the prioritization of information processing for goal-oriented behaviour, which is especially important under demanding, ambiguous, or novel conditions (Posner & Snyder, 1975). Building on early lesion studies (Stuss & Benson, 1984; Trimble, 1986), Stuss and colleagues (Stuss et al., 2007; Stuss, 2011) proposed a model of frontal lobe functions in which executive control arises from interactions between various functions of attention, namely *energization* and a set of *executive functions* including *task-setting* and *monitoring*. They also designed a set of paradigms, grouped in the ROTman-Baycrest Battery to Investigate Attention,

ROBBIA^B (Stuss et al., 2005), that enable the exploration of these functions in focal brain damaged patients during the chronic stage of recovery. *Energization* refers to the production of an optimal level of arousal and readiness to respond, initiate, and sustain activity (Aston-Jones & Cohen, 2005; Sturm & Willmes, 2001; Stuss & Benson, 1984). This function was termed in other cognitive models as the interaction between *tonic* and *phasic*^G alertness (Aston-Jones & Cohen, 2005; Godefroy & Stuss, 2007; Sturm & Willmes, 2001; Stuss & Benson, 1984), or the *alerting* function of attention (Posner & Petersen, 1990). Deficits of energization present in neuropsychological evaluations as longer Reaction Times (RT, see task details in Box 1, henceforth indicated by ^B) in simple target detection and choice tasks, difficulty sustaining preparation to respond, fewer responses in verbal fluency^B tests, and inefficient Stroop test^B performance (Stuss & Alexander, 2007). *Task-setting*^G involves selection of a stimulus–response relationship that is important for the planning of an upcoming task (Stuss & Alexander, 2007). Task-setting also involves the maintenance of task-related representations in order to guide behaviour (e.g., Guided activation model; Miller & Cohen, 2001), thus linking this function and working memory. A recent neurocognitive model of task setting was proposed by Domenech and Koechlin (2015), which defines *task sets*^G as executive units that are formed by experience and stored for efficient adaptive behaviour in everyday environments. According to this model, interaction with the environment depends on a continuous interplay between adjustments of previously learned task sets (i.e., *exploiting*), and the creation of new ones (i.e., *exploring*). More generally, impairments in task setting lead to the formation of defective stimulus–response contingencies (Godefroy et al., 1994, 1998), resulting in an increased number of errors (e.g., false positives) in neuropsychological tests such as verbal fluency^B, the Wisconsin Card Sort Test (WCST)^B, or the Trail Making Test (TMT)^B (Stuss & Alexander, 2007). *Monitoring* refers to the executive control of performance and aims to examine if the ongoing task is being performed adequately. Impairment in monitoring may manifest as different types of errors, such as increased false positives, false negatives, omissions, or failure to show appropriate changes in RT or to note that an error has occurred (Stuss & Alexander, 2007). The monitoring function has been the focus of investigations of brain correlates of conflict monitoring (Badre & Wagner, 2004; Botvinick et al., 2001; Botvinick et al., 2004a; Yeung, 2013), conflict adaptation (Mayr & Awh, 2009), error likelihood and prediction models (Brown & Braver, 2006, 2009). Monitoring is related to the *updating* function, which is included in other influential theories of executive functions (Miyake & Friedman, 2012; Miyake et al., 2000). Yet, the extent to which these different functions involve the same behavioural and neural mechanisms is still debated.

Box 1 Principal neuropsychological tests used to assess attention, working memory, and executive control. RT: reaction time.

Neuropsychological test	Description (see for details MacPherson & Della Sala, 2015)
Delayed-response task a. non-human primate (Finan, 1942; e.g., Jacobsen, 1935) b. human (e.g., Freedman & Oscar-Berman, 1986; Rodriguez & Paule, 2009)	Task: Extensively used in the non-human model, in this task participants are instructed to recall previously presented information after a delay of several seconds Versions: A variety of versions has been used in the literature, including the Simple Delayed Response Task, the Delayed Alternation Task (e.g., Mishkin, 1957), the Delayed Match to Sample Task (e.g., Bauer & Fuster, 1976), Delayed Non-Match to Sample Task (e.g., Mishkin & Delacour, 1975); variants use spatial stimuli, objects (shape or colour), face stimuli Response strategy: Task completion requires maintaining information stored in working memory during the delay Measures: Proportion of correctly recalled items, often followed by the presentation of a feedback (e.g., in the form of food in studies using non-human primates)
Digit span WAIS-IV (Wechsler, 2008)	Task: Participants are instructed to repeat a series of digits read by the examiner at a rate of one digit per second Versions: (1) Digit span forwards (participants repeat digits in the same order); (2) Digit span backwards (participants repeat digits in the reverse order); (3) Digit span sequencing (participants repeat digits in order from the lowest to highest number) Response strategy: Both versions require short-term memory. Digit span backwards also requires manipulation within working memory (Baddeley, 1992) and it is thought to measure executive abilities (Robbins, 1996) Measures: Proportion of correctly recalled sequences, the maximum sequence length recalled, a discrepancy score (the difference between Digit span forwards and Digit span backwards)
Executive RT test (Hartikainen et al., 2010)	Task: Participants are required to report the orientation of a triangle in each trial. The go/no-go signal is a traffic light where go/no-go is either a green or red light and changes after every block throughout the experiment with either a neutral (flower) or an emotional (spider) distractor present conditions Response strategy: Participants press, or withhold from pressing, a corresponding response button according to go/no-go signal Measures: RTs and the number of basic errors (incorrect response, missed response, or commission error) and composite errors (shift errors and total errors) under different conditions provide measures of working memory, inhibition and shifting
Flanker task (Eriksen & Eriksen, 1974) This task is included in the Attention Network Test (ANT: see Fan et al., 2002)	Task: Participants respond to a central letter or object, ignoring the conflicting stimuli to either side (related to the alternative response) Conditions: Trials can be: (1) congruent: both central and side stimuli require the same response or (2) incongruent: two conflicting responses are elicited by central and side stimuli Response strategy: Suppress the automatic tendency to respond to the task-irrelevant dimension (side stimuli) and use the task-relevant dimension in order to guide behaviour (central object) Measures: Accuracy and RTs. Significantly greater RTs are observed for incongruent trials compared to congruent trials (i.e., the incongruity effect) or on the subsequent trial following an incorrect response compared to correct responses (i.e., post-error slowing)
Line bisection (Schenkenberg et al., 1980) This test is included in different test batteries assessing neglect (see the battery of GEREN, (Azouvi et al., 2006) and the Behavioral Inattention Test (BIT: Wilson et al., 1987)	Task: Participants are instructed to bisect a horizontal line as accurately as possible Versions: (1) Long lines (20 cm); (2) Short lines (5 cm); (3) Landmark (i.e., pre-bisected lines; participants indicate the longest or shortest segment) Response strategy: Participants generally compare the length of segments. Normal participants may bisect lines to the left of the veridical center (pseudoneglect phenomenon, see Bowers & Heilman, 1980; Toba et al., 2011); neglect patients typically err rightwards, considering the right segment longer than the left one Measures: Deviation from the veridical center (in mm) or percentage of deviation. Worsening of performance associated with neurological conditions is generally observed as a greater rightward deviation from the veridical center
N-back task See Smith et al. (1996) for the original task	Task: Participants see a sequence of items (e.g., numbers, letters), each item being followed by a blank delay interval. Participants decide whether the current item is the same as the one that occurred <i>n</i> items earlier in the sequence (e.g., two items for a 2-back task) Conditions / Versions: Task difficulty can be manipulated by increasing the interval (number of interceding stimuli) between the target and the probe (2-back, 3-back ... <i>n</i>-back) Response strategy: For accurate responses, each presented item must be maintained in the sequential order of presentation (working memory load), all items being potential targets Measures: Accuracy and RT. Worsening of performance is observed for higher levels of working memory load

Neuropsychological test	Description (see for details MacPherson & Della Sala, 2015)
Reaction time (RT) tests (Deary & Stough, 1996; Luce, 1986). This type of task has different versions and can be either included in different test batteries (Godefroy et al., 1994, 2010) or presented as an independent test	Task: Participants are instructed to respond as rapidly and as accurately as possible Versions (examples): (1) Simple RT test (participants respond by pressing a button to stimuli displayed with randomly varying stimulus onset asynchronies); (2) Choice RT test (participants indicate which stimulus has been displayed by pressing one of two response-keys) Response strategy: Different, depending on the version used. Motor, perceptive, attentional and decision processes can be inferred Measures: Accuracy (errors and responses occurring outside the range [anticipations and omissions]) and RTs (RT indices [e.g., minimum RT, i.e., the fastest RT] and mean RT are analysed)
ROBBIA (ROTman-Baycrest Battery to Investigate Attention: Stuss et al., 2005)	Task: Battery composed of 11 subtests (Simple RT, Choice RT, Prepare RT, Concentrate, Count, Divide, Tap, Switch, Nogo, Suppress, Set) assessing different supervisory processes (energizing, monitoring, adjustment of contention scheduling, inhibition, task setting, control of logic) Versions: In later versions, Stuss and Alexander (2007) considered that “inhibition” could be explained by energization, task setting and monitoring and thus it was not included in the supervisory processes Response strategy: Different depending on the subtest; each subtest involves different attentional supervisory processes Measures: Accuracy and RT, intraindividual standard deviation (for Tap), false alarms. Contrasted performances can be observed for the different tests of the battery because they reflect different attentional supervisory processes. Interpretation of behavioural data and clinical anatomical correlations involves assembling related tasks, analysing the underlying supervisory processes and correlating deficits in the tasks with lesions localization
Simon task (Simon & Rudell, 1967; Simon & Small, 1969)	Task: Participants respond to a nonspatial attribute (e.g., colour) of a lateralized target irrespective of the spatial position of the target stimulus; spatial compatibility between the position of the target (left or right) and the position of the response (e.g., left or right key press, the colour of the stimulus being assigned to a certain response) is manipulated Conditions: Trials can be either (a) congruent, in which both stimulus features (e.g., position and colour) require the same response or (b) incongruent, in which two conflicting responses are elicited by stimulus features Response strategy: Suppress the automatic tendency to respond to the task-irrelevant dimension (position) and use the task-relevant dimension in order to guide behaviour (colour) Measures: Accuracy and RTs. Significantly greater RTs are observed for incongruent trials compared to congruent trials (i.e., the incongruency effect) or on the subsequent trial following an incorrect response compared to correct responses (i.e., post-error slowing)
Stroop test (Stroop, 1935)	Task: Participants respond to lists of words, based on the colour of the ink in which a word is printed, ignoring the semantic information Conditions: Trials can be (a) congruent, in which both the word and the ink colour prompt the same response, or (b) incongruent, in which two conflicting responses are elicited by the word and the ink colour Response strategy: Suppress the automatic tendency to respond to the task-irrelevant dimension (semantic) and use the task-relevant dimension in order to guide behaviour (ink colour) Measures: Accuracy and RTs. Significantly greater RTs are observed for incongruent trials compared to congruent trials (i.e., the incongruency effect) or on the subsequent trial following an incorrect response compared to correct responses (i.e., post-error slowing)
Trail Making Test (TMT) Initially developed for military screening in 1944; this test is included in Halstead–Reitan Neuropsychological Test Battery (Reitan & Wolfson, 1985, 1993) and the Delis–Kaplan Executive Functions System (D-KEFS) battery (Delis et al., 2001)	Task: This test consists of two parts: (a) participants are instructed to draw a trail connecting numbered circles in ascending numerical order as quickly and as accurately as possible, without lifting their pencil from the paper; and (b) participants are required to draw a trail connecting the numbers and letters in ascending numerical and alphabetical order, switching between the numbers and letters Versions: Trail Making Test from the D-KEFS includes four baseline conditions (i.e., visual scanning, number sequencing, letter sequencing, and motor speed) in place of Part A Response strategy: Completion of Task A requires a good motor and visual search speed. The difference between the time to complete Part A and Part B is used as a measure of set shifting Measures: Time taken to complete each part, and difference between completion times corresponding to Part A and Part B, number of errors

Neuropsychological test	Description (see for details MacPherson & Della Sala, 2015)
Verbal fluency test (Borkowski et al., 1967) This task is included in different test batteries assessing: (1) executive functions (e.g., Frontal Assessment Battery: Dubois et al., 2000); the Delis–Kaplan Executive Function Scale (Delis et al., 2001) and (2) language abilities (e.g., Multilingual Aphasia Examination: Benton et al., 1994)	Taks: Participants are instructed to generate as many words as possible in a predefined time (usually one minute), while avoiding producing proper nouns or repetitions. Some versions also explicitly prohibit numbers Versions: (1) Letter-cue (participants generate words with a particular letter; “F”, “A”, and “S” are the most widely used), and (2) semantic (participants generate words belonging to a given category, e.g., animals, vegetables, occupations) Response strategy: Letter-cue fluency involves strategic search through lexical memory. Semantic fluency involves search through semantic memory (failures may be due to impairments in language or semantic memory rather than executive control) Measures: Total number of words produced, number of repeated words (perseverative errors), cluster length of words from the same letter-cue or semantic subgrouping, and number of category switches made
WAIS-IV (Wechsler Adult Intelligence Scale, 4th edition) (see Duncan, 2013 for review; see also Spearman, 1927 for the <i>g</i> factor; Wechsler, 1981 for the WAIS-R)	Task: A measure of intellectual functioning comprising a core of 10 subtests (plus additional optional subtests) tapping verbal comprehension, perceptual reasoning, working memory, and processing speed. Overall, Full scale IQ (Intelligence Quotient (IQ) and General Ability Index (GAI, a measure of overall ability less dependent on working memory and processing speed) scores as well as index scores reflecting the 10 subtests are calculated. Subtests encompass nonverbal (e.g., matrix reasoning, picture completion, digit-symbol coding) and verbal (e.g., vocabulary, similarities, arithmetic, digit span) subtests measuring fluid and crystallized intelligence. Thought to reflect the <i>g</i> factor (or general intelligence, see also Glossary scales) Versions: Revised WAIS (WAIS-R), WAIS-III, WAIS-IV Response strategy: Different depending on the subtest. Taps aspects of fluid and crystallized intelligence. It generally involves major components of intelligence (verbal comprehension, perceptual reasoning, working memory, processing speed) Measures: Accuracy and processing speed, depending on whether the subtest is timed or untimed. RT; a greater IQ score results from higher levels of performance to each subtest
Wisconsin Card Sorting Test (WCST) (Grant & Berg, 1948)	Task: Participants are required to classify a set of cards based on one of three different criteria (colour, shape, or number of symbols in the card). Participants are not told the rule set by the experimenter instead they are required to discover it through trial and error; the rule changes without participant’s knowledge Versions: Original (Grant & Berg, 1948), modified version (Nelson, 1976); 64-card WCST (Greve, 2001); for a similar sort task, see Weigl Color–Form Sorting Task (K. Goldstein & Scheerer, 1941; Weigl, 1927) Response strategy: Task completion requires to adapt behaviour based on the feedback received, and adjust accordingly without persevering on a rule when notified about the incorrect classification Measures: At each trial, participants receive feedback on whether their classification was correct or incorrect. The experimenter counts the number of contingencies (sorting rules) achieved, the sum of perseverative (when participants keep applying the old rule) and non-perseverative (mistakes made when in trials in which the rule changed) errors, total correct card sorts, number of trials taken to achieve the initial sort rule

Taken together, the cognitive models discussed above represent only a portion of perspectives on executive control, and depict a highly fragmented conception of differing components, ultimately sharing some commonalities. The function termed inhibition^G is exemplary in this case, since this function has received a great deal of coverage in the literature due to the widespread use of the Stroop test^B in studies of executive control. However, the conceptualization of inhibition in executive control models is debated, and has been associated with attention (e.g., Gaspelin & Luck, 2018), working memory (Redick et al., 2007), and cognitive control (Miyake et al., 2000). Miyake and colleagues (Miyake et al., 2000) found that the functions of inhibiting^G of prepotent responses, shifting between mental sets or tasks, and updating and monitoring of working memory contents, were moderately correlated with one another but also separable, and therefore suggested that there is both diversity and unity among these functions.

In an effort to synthesise the distinct theoretical frameworks responsible for the unstructured growth described above, attempts were made to capture the extent of overlap

between these functions (e.g., Chen et al., 2019; Miyake et al., 2000). While the uniqueness of each function is still debated (Bae & Luck, 2018; Tas et al., 2016), both behavioural and anatomical evidence point to some overlap between functions (Awh et al., 2006; Jonikaitis & Moore, 2019; Theeuwes et al., 2009). Below, we critically evaluate the anatomical substrates and functional correlates of these different functions to gain insight into the relationship between the different facets of these processes.

Lesion Studies

Descriptions of neuropsychological impairments resulting from frontal lobe injuries have proven paramount in the understanding of functional and anatomical correspondence between attention, working memory, and executive control (for example, see Burgess et al., 2007; Heilman & Watson, 1977; Knight, 1991; Luria, 1973; Mesulam, 1985; Mirsky et al., 1991; Norman & Shallice, 2000; Posner & Petersen, 1990; Robbins, 2007; Shallice, 1982; Stuss & Alexander,

2000). These neuropsychological impairments are often grouped under the term *dysexecutive syndrome* (see Baddeley & Wilson, 1988), referring to the presence of behavioural or cognitive disorders related to executive functions (see for details Godefroy et al., 2018), regardless of the lesion site (Anderson et al., 1991; Vataja et al., 2003). This term is conceptually distinct from the broader term “frontal lobe disorders or syndromes”, which refers to a topographic syndrome that groups all functional disorders together when a frontal lesion is present, without considering that executive functions involve different (i.e., not only frontal) neural systems. The diversity of neural systems and processes grouped in the dysexecutive syndrome raises the question of whether this syndrome should be differentiated (i.e., considering deficits of different executive functions tested with different neuropsychological tests as distinct entities) or unitary (see for instance Stuss & Alexander, 2007). Different cognitive models highlight a central role of both attention and working memory in executive control, however, attention and working memory can diverge following certain lesions which cause distinct deficits in components of attention and working memory. Below we review this evidence, which may aid in distinguishing the extent to which executive control is fractionated.

The lesion studies pioneered by Stuss and colleagues included patients with various etiologies (e.g., stroke, tumor, trauma) to explore the dysexecutive syndrome of focal brain damaged patients in the chronic stage of recovery (see also Stuss et al., 2005), by avoiding the bias introduced in clinical anatomical correlations by the vasculature of the brain (see Godefroy et al., 1998; Ivanova et al., 2020; Mah et al., 2014; Toba et al., 2020a, b). Stuss and Alexander (2007) mapped the distribution of frontal lesions in 38 patients and compared the performances of patients with and without damage in a particular brain region. Results revealed a limited number of independent frontal or control processes called “branches” (i.e., energization, task setting, and monitoring) with distinct anatomical underpinnings that could explain performance in each task.

First, deficits in energization were described following lesions of bilateral (but with a right predominance) of dorsomedial prefrontal cortical (DMPFC) regions including the anterior cingulate cortex (ACC) and presupplementary motor area (preSMA, Fig. 1). This result added more evidence to the slowing and increased response variability observed following DMPFC lesions in tasks requiring executive control (Shallice et al., 2008; Stuss et al., 2005). Shallice et al. (2008) also reported that the ACC, particularly in the right hemisphere, was important for energization. Furthermore, Posner and DiGirolamo (1998) specified that in this network, the ACC was essential for tasks involving error detection, novelty, or conflict.

Second, Stuss and Alexander (2007) also observed task setting impairment following damage to the left lateral prefrontal cortex, primarily in the ventrolateral prefrontal cortical region (VLPFC, Fig. 1). This is consistent with early evidence showing that patients with lesions of the left frontal lobes were unable to set and apply a stimulus–response relationship, despite being able to comprehend the meaning of instructions (Luria, 1966). Random errors in *rule learning*, *task switching*, and *shifting between attentional sets* (i.e., shifting between tasks requiring different stimulus–response pairings) that might result from transient lapses of attention (Barceló & Knight, 2002) have also been reported in patients with left dorsolateral prefrontal cortex (DLPFC) lesions.

Third, the Stuss and Alexander (2007) model linked monitoring impairment to lesions of the right DLPFC (Fig. 1) and observed that patients with such lesions did not show the “foreperiod effect” observed in control participants (i.e., decreasing of RT when the interstimulus interval increases). Hence, the DLPFC was considered an important node for the time-monitoring process (Stuss et al., 2005). These findings have been further replicated in Vallesi et al. (2007), where the role of the right DLPFC was associated with time monitoring supervisory function. Further, Funderud et al. (2013) showed that the DLPFC is also important in *preparatory attention* (i.e., getting ready for a task following a warning signal). *Selective spatial attention* impairments have been linked to lesions of both the DLPFC and VLPFC in patients presenting *neglect^G* attentional impairments (Husain & Kennard, 1996; Ptak & Schneider, 2010; Stone et al., 2011; Toba et al., 2017, 2018a; Verdon et al., 2010), a clinical condition involving both non-executive and executive components (see Husain, 2019 for a review). These impairments include a directional bias in the *orienting of attention* towards the ipsilesional side (Karnath, 2015) and in the competition for *selection* (Gainotti et al., 1991; Toba et al., 2018b), an impaired *selective attention* at fixation (Husain et al., 1997), a *sustained attention* deficit (Malhotra et al., 2009; Robertson et al., 1995) and a spatial *working memory* deficit (Malhotra et al., 2005; see also Moreh et al., 2014 for different findings). However, based on neuropsychological observations, Shallice et al. (2008) concluded that lesions of the right VLPFC reduced monitoring capacity but did not impair general alerting or energization functions. The role of all these anatomical regions in attention and cognitive control has been proposed also in other cognitive models, reviewed by Szczepanski and Knight (2014).

Disorders of the working memory system have also constituted an important window to advance our understanding of cognitive control (Baddeley, 2012; Fuster, 2001; Miyake et al., 2000). The working memory system was initially analysed in animal studies showing that the DLPFC is involved in the temporary storage of information during delayed-response tasks (Goldman-Rakic, 1987; Jacobsen, 1936).

Later, the more general dysexecutive syndrome was linked to the *central executive* component of working memory in humans (Baddeley & Hitch, 1974), with dysfunction often resulting in impairments in task coordination, inhibition^G, switching, and updating (Baddeley, 1996; Miyake et al., 2000). According to other authors (Roussel et al., 2012), working memory impairment cannot be the core feature of executive disorders, even though it does contribute to some of these disorders.

Evidence exists to show that working memory processes in humans are also subserved by the DLPFC, as lesions of this region are associated with impairments to the maintenance, monitoring, and manipulation of working memory content, and working memory related deficits in planning and problem solving result in omissions, perseverations, and performing actions in an incorrect order (Duncan, 1986; Grafman, 1989). Different tasks have been used to assess working memory skills in brain-damaged patients. First, *delayed-response task*^B impairments in neurological patients have been associated with lesions of the DLPFC (see D’Esposito & Postle, 1999 for a meta-analysis). Second, the assessment of working memory with the *digit span*^B backward task shows that patients with unilateral right (Heilbronner et al., 1991), left or bilateral frontal lobes lesions (Stuss et al., 2001) have difficulties in performing this task. This includes both DLPFC lesions (D’Esposito and Postle (1999) and medial prefrontal lesions (Leimkuhler & Mesulam, 1985; Wallesch et al., 1983), but there is also evidence that a decrease in digit span performance can occur following subcortical lesions of the caudate (Parkin et al., 1996) and the thalamus (Van Der Werf et al., 1999). Third, impairments in the *n-back task*^B have been associated with lesions of both the DLPFC and VLPFC (see Müller et al., 2002). In a small sample of patients who had undergone resection

for brain tumors centered in the left superior frontal gyrus (SFG), reduced performance relative to controls was found across spatial, facial, and letter versions of the n-back task, with the left DLPFC (but not the DMPFC) seeming to be an essential component of the working memory network, regardless of verbal, facial, or spatial task material (du Boisgueheneuc et al., 2006). Also using an n-back task, Tsuchida and Fellows (2009) reported impaired performance in patients with DLPFC and VLPFC lesions in the left hemisphere but also in patients with DMPFC lesions including the ACC. By contrast, patients with lesions in the right LPFC or orbitofrontal cortex (OFC) regions showed no decline in performance on this task. Based on their observations, these authors also suggested that the lateral portion of the PFC may be involved in the working memory aspect of this task.

Verbal fluency, Wisconsin Card Sorting Test (WCST), and Trail Making Test (TMT) tasks have been used in lesion studies to investigate the anatomical correlates of executive functions such as initiation, inhibition, and mental flexibility (within the Stuss & Alexander, 2007 model, all of these functions rely principally on task setting, but involve monitoring and energization, although the role of the working memory is also clear). However, other studies suggest that the DLPFC is not critical to perform these tasks. Using WCST, Stuss et al. (2000) observed that patients with DLPFC damage presented with fewer sorting categories and more perseverative errors than other individuals with lesions, although patients with VLPFC lesions also committed perseverative errors. A recent study using voxel-based lesion-symptom mapping (VLSM^G) confirmed that right VLPFC lesions (in addition to more posterior brain regions, as well as substantial involvement of underlying white matter, which was not examined) was associated with increased

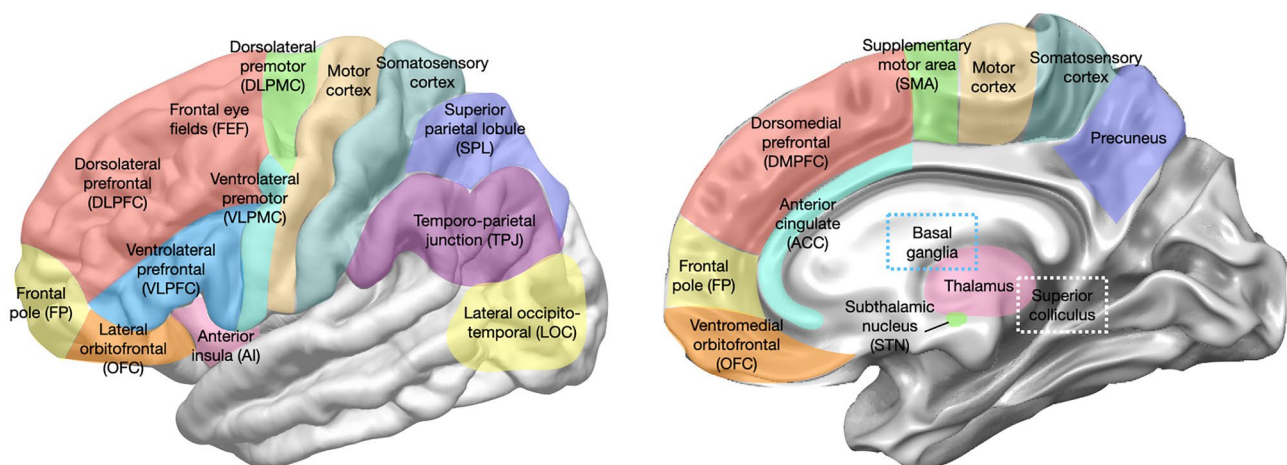


Fig. 1 Widespread brain regions are recruited during attention, executive control, and working memory tasks, including the basal ganglia and superior colliculus (dashed boxes—not visible in these views)

perseverative errors on the WCST^B (Gläscher et al., 2019). Other studies reported that patients with damage to both the DLPFC and ACC exhibited an increased rate of perseverative, non-perseverative, and efficiency (shifting away from the correctly attained sorting rule) errors (Barceló & Knight, 2002). Demakis (2003) observed that patients with frontal lesions performed more poorly than patients with posterior lesions, but that the lesion location did not predict test performance. Some lesions studies suggest a lateralization in WCST performance with left frontal lobe damage increasing the perseverative error rates (i.e., repeating a rule that is no longer appropriate) and reducing the number of categories successfully sorted (Drewe, 1974; Goldstein et al., 2004), although other studies did not find any lateralization (Nelson, 1976).

Using the TMT, Stuss et al. (2001) compared the performance of patients with frontal lesions with those of patients with non-frontal lesions and of healthy controls. They also observed that patients whose performance was significantly impaired relative to controls had DLPFC lesions, whereas patients with inferior medial frontal areas lesions made the fewest errors. Yochim et al. (2007) later confirmed these findings, reporting that patients with DLPFC lesions were disproportionately slower than controls on the TMT Part B. By contrast, the VLSM study of Glascher et al. (2012) reported that lesions of the rostral ACC, but not in the DLPFC, were associated with significantly poorer performance. Additionally, lesions in the cerebellum have also been reported to result in poorer TMT performance compared to controls (Alexander et al., 2012; Exner et al., 2004; Gottwald et al., 2004). Concerning the lateralization of structures underpinning performance on this test, it has been reported that left frontal damage resulted in more errors and longer completion times compared to the performance of healthy controls (Gouveia et al., 2007) and of either left or right non-frontal patient groups (Hoerold et al., 2013). By contrast, Gottwald et al. (2004) found that lesions to the right, but not left, side of the cerebellum resulted in slower completion on Parts A and B.

Performance in stimulus–response compatibility tasks (e.g., Stroop^B, Simon^B, and Flanker tasks^B) is thought to reflect both attentional and working memory functions interacting for executive control. In lesion studies, impairments in task performance are associated with DLPFC damage (Perret, 1974; Stuss et al., 2001; Vendrell et al., 1995), although this region does not seem to be specific to incongruency and error (Floden et al., 2011). Lesions of the DMPFC have also been associated with longer RTs and increased errors in incongruent trials (Richer et al., 1993; Stuss et al., 2001). Furthermore, lesions of the ACC are associated with longer RTs in incongruent trials (Cohen et al., 2008; R. A. Cohen et al., 1999; Ochsner et al., 2001; Swick & Turken, 2002), although this has not been confirmed by other studies (see

Fellows & Farah, 2005; Swick & Jovanovic, 2002). A number of studies have also highlighted a critical role for the right VLPFC, specifically the inferior frontal gyrus. VLSM^G using post-operative resection cavities in right frontal lobe brain tumour patients linked localised lesions of this region to longer response times on the Stroop task (Puglisi et al., 2019), and similar results have been demonstrated using the stop-signal task (Aron et al., 2003, 2014).

The Stuss and Alexander (2007) model was later revised to include *behavioural* or *emotional self-regulation* (Levine et al., 2000; Stuss, 2011). Studies investigating anatomical correlates of *attention for emotional stimuli* in neurological patients reported that these impairments were linked to lesions of the OFC and the frontal pole (Fig. 1, Hartikainen & Knight, 2003). Lesions in the ventromedial orbitofrontal cortex (VMOFC) and frontal pole have also been associated with value-based decision making and impulsivity (Berlin & Rolls, 2004; Gläscher et al., 2012; Noonan et al., 2017). According to the *gateway hypothesis*, the DLPFC plays an important role in the *supervisory attentional gateway* system that enables switching between attending modes, either to the outside world or to the inner mental world (see Burgess & Wu, 2013).

To sum up, anatomical elements presented in this section sustain both the unity and diversity of attention, working memory, and cognitive control. Different studies have emphasized selective impairments of these functions and different anatomical correlates within the frontal lobes (Godefroy et al., 1999; Shallice, 2002; Stuss et al., 1995), a concept also confirmed by studies reporting double dissociations in patients with prefrontal cortex lesions (Camille et al., 2011; Shamay-Tsoory et al., 2009; Tsuchida & Fellows, 2009). However, neuropsychological evaluations also demonstrate that one or a few *core* cognitive deficits (often underlined by similar lesions) are able to explain impairments in complex tasks requiring multiple cognitive operations for successful completion (see also Szczepanski & Knight, 2014). Given the variety of impairments related to these processes, we argue here that attention and working memory are core cognitive operations that flexibly interact in order to produce efficient cognitive control. By their nature, lesion studies can only determine which regions are crucial for a function to be performed, but not all regions that are involved in its processing, which is made possible by non-invasive neuroimaging, in particular functional MRI which we discuss in the following section.

Neuroimaging Evidence

Functional MRI Studies

Functional magnetic resonance imaging (fMRI)^G studies reliably show overlap in the brain regions recruited during

tasks probing attention, working memory, and cognitive control, and have highlighted that although there may be functional overlap, there is also a degree of anatomical dissociation. Both task-based and resting state fMRI findings indicate that there are two key networks recruited during these processes, including a set of fronto-parietal cortical regions (Cole & Schneider, 2007; Duncan, 2013) and a separate set of cingulo-opercular regions (Dosenbach et al., 2008; Power et al., 2011), also recruiting the thalamus and basal ganglia (Fig. 2). Resting state fMRI^G studies (Hwang et al., 2017) have also suggested that thalamo-cortical connectivity might play a role in interactions between cognitive functions underlying cognitive control. Specifically, it has been proposed that the thalamus mediates interactions between attention and working memory (de Bourbon-Teles et al., 2014; Saalman et al., 2012; Zhou et al., 2016) and seems to have an important role in integrating functions (Fiebelkorn & Kastner, 2019). Integrating a temporal dimension, the study of Dosenbach et al. (2008) suggested that the fronto-parietal network provides signals important in initiating and adjusting control on a rapid time scale, whereas the cingulo-opercular network provides signals that allow set maintenance over a longer time scale.

Two fronto-parietal networks, the dorsal (DAN)^G and the ventral (VAN)^G attention networks, have been linked to the selective aspect of attentional control in an influential model based on evidence from both task-based and resting state fMRI studies (Corbetta & Shulman, 2002; Corbetta et al., 2008). The DAN^G recruits bilateral frontal (DLPFC) and parietal regions (SPL) and supports goal-directed attention. On the other hand, right-lateralized^G VLPFC and parieto-temporal (temporo-parietal junction, TPJ) areas form the VAN, a reorienting network that consistently activates when attention needs to be diverted towards novel and salient stimuli, and collaborates with the DAN^G to implement response selection. Almost 20 years since this model was first proposed, studies consistently show that there is a high degree of interaction between these two networks, supporting the flexible allocation of attentional resources in multisensory and dynamic environments across species (Bowling et al., 2020; Callejas et al., 2014; Chica et al., 2013; Fiebelkorn & Kastner, 2020; Patel et al., 2015; Vossel et al., 2014).

Beyond brain networks supporting selective attention, distinct patterns of activation measured using task fMRI have been associated with each attentional function in the Posner and Petersen (1990) model. For instance, early investigations of the alerting function (see Attention Networks Model^G), showed the recruitment of regions of right frontal and parietal cortex as well as the locus coeruleus and thalamus (Aston-Jones & Cohen, 2005; Fan et al., 2005; Périn et al., 2010; Sturm & Willmes, 2001; Xuan et al., 2016). More recently, this function has also been linked to

the activity and connectivity of fronto-striatal (Chica et al., 2016; Xuan et al., 2016) and cingulo-opercular networks (Coste & Kleinschmidt, 2016; Dosenbach et al., 2008; Han et al., 2019; Haupt et al., 2019; Sadaghiani & D'Esposito, 2015). The orienting of attention (see Attention Networks Model^G) involves parietal, frontal eye fields, thalamic, and mesencephalic regions (Corbetta & Shulman, 2002; Corbetta et al., 2008; Fan et al., 2005; Xuan et al., 2016). The executive control network (see Attention Networks Model^G) is associated with activation of the DMPFC including the ACC, but also anterior insular (AI) cortex, and part of the basal ganglia (Berger & Posner, 2000; Braver & Cohen, 2000; Fan et al., 2005; Ott & Nieder, 2019; Xuan et al., 2016; see also Fig. 1 and Fig. 2).

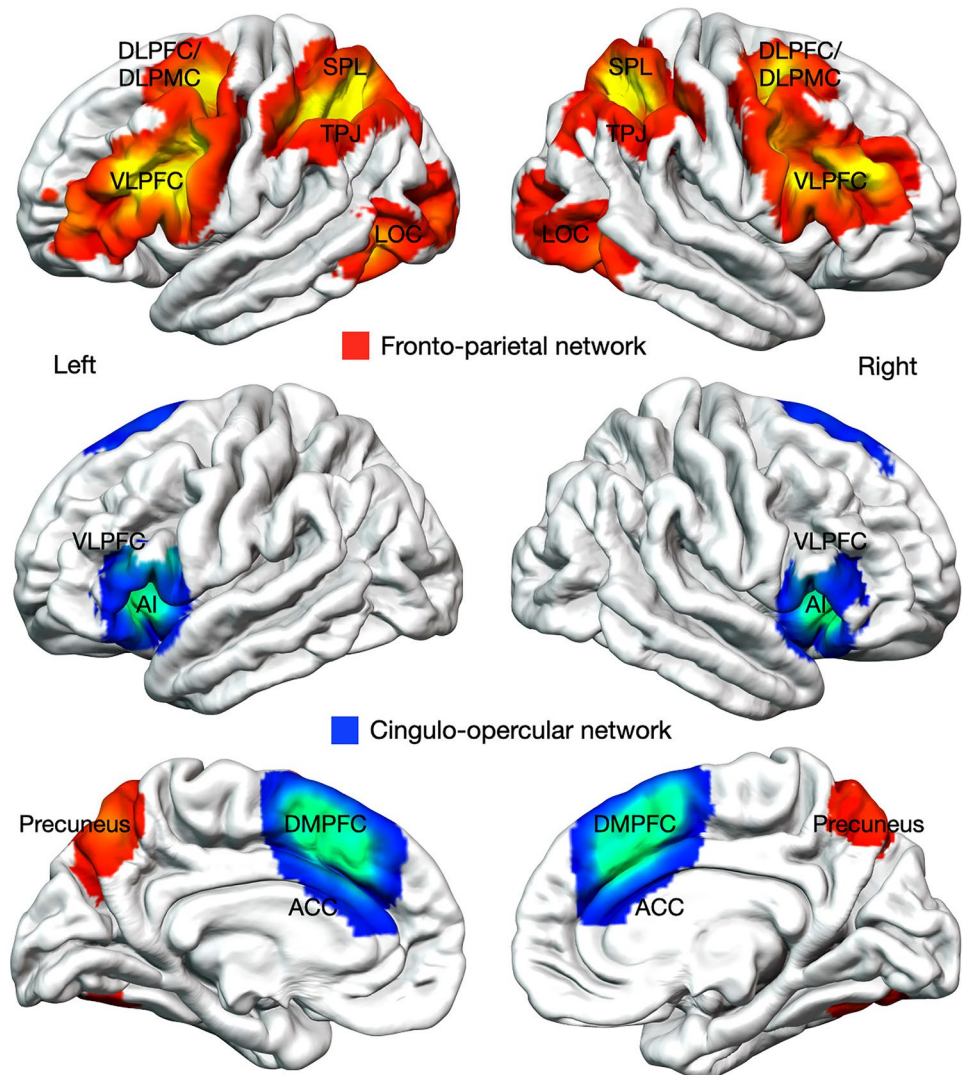
Neuroimaging studies exploring cerebral areas related to Baddeley's (Baddeley, 1996) central executive system of working memory (Collette & Van der Linden, 2002) have also demonstrated the involvement of frontal and parietal cortex in this function. Other neuroimaging studies (Collette et al., 2005) have adopted the framework proposed by Miyake et al. (2000) to explore working memory. Using Positron Emission Tomography (PET), Collette et al. (2005) observed that inhibition, switching, and updating activated common cerebral areas such as the left superior parietal cortex, the right TPJ, and the left VLPFC. In addition to these common areas, each one of the executive processes proposed in the model also relied on specific sets of frontal and parietal areas. Furthermore, a meta-analysis conducted in studies assessing working memory using n-back^B tasks (Owen et al., 2005) emphasized that task performance recruits widespread fronto-parietal areas including premotor cortex (DLPFC and SMA), prefrontal cortex (DLPFC and VLPFC), the frontal pole, precuneus, superior parietal cortex, and the TPJ (see Fig. 1 and Fig. 2). Multivoxel pattern analysis (MVPA^G) has also shown that while PFC encodes more abstract representations of the object category, ventral occipitotemporal regions also play an important role in visual working memory, encoding familiar objects such as faces, houses, scenes, bodies, and motion (Ester et al., 2013; Han et al., 2013; Lee et al., 2013; Nelissen et al., 2013; Riggall & Postle, 2012; Sreenivasan et al., 2014). As working memory tasks recruit such widespread brain regions (Miller et al., 2018), this may support state-based models suggesting that it is a property that emerges as a consequence of other functions (D'Esposito & Postle, 2015; Postle, 2006). In particular, this evidence suggests that while posterior cortex maintains feature and stimulus-specific representations, the PFC likely represents higher-order information, such as task rules, goals, or abstract representations of categories, and exerts top-down control that modulates the representation of the information maintained in posterior cortex (D'Esposito & Postle, 2015; Lara & Wallis, 2015).

The association between executive control functions and the activation of the ACC is one of the most frequently observed findings in the neuroimaging literature, as shown in studies (Diamond, 2013; Fan et al., 2003; Shenhav et al., 2016; Wu et al., 2018) that used a variety of executive control measures including the Stroop test^B (Stroop, 1935), the Flanker task^B (Eriksen & Eriksen, 1974), and the Simon task^B (Simon, 1990: see also Box 1). For example, Sarazin et al. (1998) reported significant correlations between activation of the DLPFC and ACC and WCST performance in patients with various frontal lesions including the DLPFC, VLPFC, and VMPFC. The ACC and SMA regions have been also implicated in *error detection* (Holroyd & Coles, 2002), *conflict monitoring* (Botvinick et al., 2004b; Xuan et al., 2016), *hierarchical reinforcement learning* (Holroyd & Coles, 2002), *executive attention* (Posner & DiGirolamo, 1998), *supervisory attention* (Norman & Shallice, 1986: see also the attentional control of action model^G), or *attention for action*

(Posner et al., 1988). ACC and AI are often co-activated as major hubs^G of the cingulo-opercular network (Fig. 2), a control network for task-set maintenance over long time periods of time and interacting with the rapid-adapting frontoparietal network (Dosenbach et al., 2008).

Functional MRI^G data has helped to show that frontal cortex also mediates different aspects of executive control in a hierarchical fashion running in an anterior to posterior (or rostro-caudal) gradient based on increasing levels of abstraction, although the type of abstraction is still debated (e.g., Badre & D'Esposito, 2009; Badre & Nee, 2018; D'Esposito & Postle, 2015). This is supported by structural MR studies indicating that cortical features may be organised to reflect this rostro-caudal organisation (Hilgetag & Goulas, 2020; Thiebaut de Schotten et al., 2017). Episodic control, whereby an action is selected based on an ongoing temporal context, also involves the anterior DLPFC and VLPFC, whereas the selection of action representations

Fig. 2 The fronto-parietal and cingulo-opercular networks recruited during fMRI tasks of attention, working memory, and executive control. Maps reproduced from a meta-analysis by (Wu et al., 2020). Acronyms are defined in Fig. 1



based on pending temporal context recruits the frontal pole. The work of Domenech and Koechlin (2015) and Koechlin et al. (2003) demonstrated that fMRI^G activation changes from posterior to more anterior regions of the frontal cortex, depending on the contextual information required to select a response (Fuster, 2001). This finding is in agreement with lesion studies suggesting that anterior subregions of the frontal cortex are associated with the higher order processing requirements of planning and selection of action (Burgess et al., 2007; Christoff et al., 2003; Ramnani & Owen, 2004). While the precise hierarchy of processing is still debated in the literature, it is broadly accepted that sensorimotor control of effectors is mediated by frontal premotor regions lying immediately anterior to primary motor regions, context-based control in the present moment is mediated more anteriorly, and control over schema representations are processed in the most anterior prefrontal regions (Badre & D'Esposito, 2009; Badre & Nee, 2018; for reviews see Koechlin et al., 2003; Ramnani & Owen, 2004).

The functional role of hierarchical prefrontal organisation has also been framed as a complementary series of decision making^G processes performed over task sets^G, rather than a hierarchy pertaining to the processing requirements themselves (Domenech & Koechlin, 2015). Different aspects of the decision-making process are hence mediated along both an anterior–posterior gradient and lateral-medial gradient within the frontal lobe. FMRI^G shows a lateral-medial distinction along which switching from exploiting previous and exploring new task sets^G can be achieved in VLPFC and DLPFC or DMPFC respectively, and an anterior–posterior distinction along which new task sets^G can be learned versus old strategies can be employed (Donoso et al., 2014). Different regions of the frontal lobe thus act together to facilitate selection of the best course of action based on the task at hand.

This conceptualisation of frontal lobe processing also links with other models of higher cognitive function. The multiple-demand (MD) system (Duncan, 2013; Woolgar et al., 2018) recruits common frontal and parietal brain regions involved during diverse cognitive tasks, particularly when achieving a goal through the assembly of a series of sub-tasks. The activity of the MD system linearly increases along with the increasing demands of the task at hand. This neurocognitive component may play a central role in supporting fluid intelligence^G and has been identified also at the neuronal level and in other nonhuman primates (Duncan, 2010). A broader cognitive control network was suggested based on recent meta-analyses (Mackie & Fan, 2017; Niendam et al., 2012; Wu et al., 2020), which examined hundreds of neuroimaging studies in the domains of attention, executive control, working memory, and decision-making^G and showed that a network composed of fronto-parietal and cingulo-opercular brain regions is recruited across all these

domains (Fan, 2014). Although some dissimilarities between the two exist, both the MD system and the cognitive control network point towards a common neural substrate supporting goal-directed behaviour that is able to promptly allocate mental resources across different sensory modalities and cognitive domains to cope with changing environmental demands.

Diffusion Tractography

The conceptual move toward approaching lesions in terms of their network-wide consequences (i.e., associationism) rather than as having discrete effects on cortical regions (i.e., localizationism) has transformed our understanding of the functional organisation of the brain (Godefroy et al., 1998; Thiebaut de Schotten & Foulon, 2018; Toba et al., 2020a, b). Over the last decade, the relevance of connections in cognitive functions has been greatly aided by the integration of structural neuroimaging techniques with fMRI, TMS, EEG and DES. This enables brain-behaviour relationships to be mapped with respect to cortico-cortical connectivity supported by white matter tracts (Toba et al., 2020b). Diffusion imaging tractography can indirectly model connections non-invasively, such that different pathways with highly stereotyped trajectories across individuals can be studied. This can be achieved in healthy individuals or patient populations by correlating interindividual differences in certain tracts with neuropsychological performance (Forkel et al., 2020; Toba et al., 2018a), or by combining functional and structural imaging approaches. Assessing the extent of disconnection caused by subcortical DES^G or brain lesions has provided the most conclusive understanding as to their role in cognitive functions (Thiebaut de Schotten & Foulon, 2018; Toba et al., 2018a for example).

Frontal and parietal regions involved in visual attention are connected bilaterally through white matter pathways that are organised in parallel branches, termed the superior longitudinal fasciculus (SLF), which are present in both monkeys (Petrides & Pandya, 1984; Schmahmann et al., 2009) and humans (Thiebaut de Schotten et al., 2012). This white matter bundle is subdivided into three major pathways (SLF I, II, and III, Fig. 3 and Table 1), based on their frontal projections (Thiebaut de Schotten et al., 2011). All branches have been linked to different aspects of visual attention (Bartolomeo et al., 2007; Doricchi et al., 2008; Lunven et al., 2015; Thiebaut de Schotten et al., 2005; Toba et al., 2018a; Toba, Zavaaglia, et al., 2020c; Toba et al., 2021, 2022).

The SLF I (Fig. 3 and Table 1) supports functioning of the DAN^G, and connects cortical regions involved bilaterally in top-down, goal-driven attention, enabling selection of sensory stimuli based on internal goals or expectations (Corbetta & Shulman, 2002; Thiebaut de Schotten, 2005; Urbanski & Bartolomeo, 2008). This tract runs anterior to

the dorsal cingulum, however is separated in its entirety by the cingulate sulcus. The cingulum (Fig. 3) plays a key role in supporting the DMN, and is involved during mind-wandering^G, autobiographical, episodic, and semantic memory (Froudust-Walsh et al., 2015; Shapira-Lichter et al., 2013). When performing a goal-directed task, the DMN is deactivated when the DAN^G is engaged, despite their close anatomical correspondence (Anticevic et al., 2012; Fransson, 2005).

No intraoperative studies have yet demonstrated interference on visual attention or executive control tasks following application of subcortical DES to the SLF I (see Sarubbo et al., 2020). However, resection of SLF I has been linked to deficits in sustained attention following neurosurgery (Howells et al., 2020) and its asymmetry has been associated with interindividual differences in visually guided control of manual actions (Budisavljevic et al., 2017; Howells et al., 2018).

The SLF II (Fig. 3 and Table 1) serves as an anatomical link between the frontal component of the DAN^G and parietal component of the VAN, and plays a fundamental role in the efficient deployment of attention. Greater right than left structural asymmetry of the SLF II has been linked to greater right hemispheric specialisation for visuospatial attention^G (using the line bisection^B test). As previously described, DES applied to the cortex of the supramarginal or angular gyri and the caudal portion of the superior temporal gyrus produces significant rightward deviation when patients perform an intraoperative line bisection^B test, demonstrating the necessity of the superior temporal gyrus in orienting attention as it relates to the voluntary exploration of space (Nakajima et al., 2017; Thiebaut de Schotten et al., 2005; Vallar et al., 2014). However, this result is more profound when DES is applied subcortically (Thiebaut de Schotten et al., 2005), also causing ocular deviation (Montemurro et al., 2016). The marked spatial neglect^G resulting from SLF II stimulation is proposed to reflect disconnection between the dorsal (DAN) and ventral (VAN) fronto-parietal networks (Bartolomeo et al., 2012).

The ventro-lateral fronto-parietal branch, the SLF III (Fig. 3 and Table 1), supports the functioning of the VAN which is also functionally right-lateralized (Corbetta et al., 2008; Thiebaut de Schotten et al., 2011). Growing evidence has led to the hypothesis that ventral regions connected by the SLF III may enable flexible reorienting between environmentally-directed processes, mediated by the DAN^G (SLF I), and internally-driven activities mediated by the DMN^G through the cingulum (Fox et al., 2006). It has been proposed that the VAN^G may generalise far beyond perceptual processing for action because of its involvement in social cognition (Catani & Bambini, 2014).

In the parietal lobe, the parietal inferior-to superior tract (PIST, Fig. 3 and Table 1) also connects medial and lateral regions. It has been speculated that these regions play

an important role in visual attention given their connecting parietal terminations of the DAN and VAN (Catani et al., 2017).

In addition to connections between different cortical regions, subcortical projections are fundamental for higher cognitive functions. Stuss and Benson (1984) proposed that prefrontal thalamo-cortical connections provide a selective gating mechanism involved in alerting and orienting processes. Studies performed in non-human primates have also shown that the pulvinar, which is in the posterior thalamus and strongly connected with visual occipital regions, is involved in coordinating functional interactions across cortical areas for visual attention (Saalmann et al., 2012; Zhou et al., 2016). This region is also connected with the superior colliculus, a midbrain structure that enables both stimulus-driven and goal-driven attention across species (Friedrich et al., 2020; Sreenivasan & Sridharan, 2019).

Branches of the SLF have also been reported as being involved in executive control (Muir et al., 2015). While no studies have yet been performed to directly test the role of the SLF II using intraoperative executive control tasks, it has been demonstrated that combining a Stroop test^B during fMRI^G with tractography highlights that the regions active during the task are connected by the SLF II, indicating the SLF II also plays an important role in executive control and conscious perception (Martín-Signes et al., 2019).

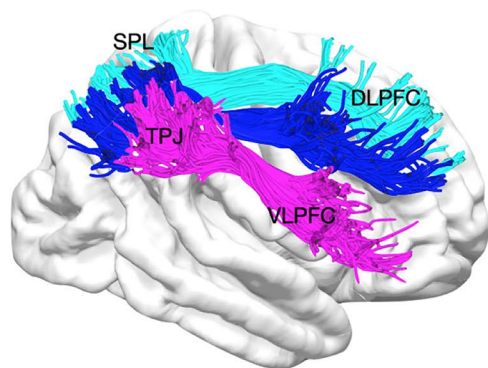
Damage to the SLF III has also been associated with post-operative declines in cognitive flexibility tested in glioma patients (Mandonnet et al., 2017). Although no intraoperative studies have tested the involvement of the SLF III in visual working memory, structural variability of this tract in healthy adults has also been linked to this function (Chechlacz et al., 2015), as well as to visuomotor processing through sensorimotor transformation (Budisavljevic et al., 2017). Puglisi et al. (2019) applied DES^G subcortically in white matter beneath the right inferior frontal sulcus, in a region sometimes termed the inferior frontal junction, causing consistent transient deficits on the Stroop test^B in 34 frontal glioma patients. They showed that this interference was caused by stimulation of the SLF III, and that resection of this tract is linked to postoperative slowing. Recent evidence has also shown that stimulation of the SLF III also causes metacognitive deficits through a transient loss of motor awareness (Fornia et al., 2020) and that its damage can cause anosognosia for hemiplegia in rare cases (Pacella et al., 2019).

Branches of the SLF connect more posterior regions with segregated regions of prefrontal cortex running medial to lateral based on gyral anatomy. Linking this anatomy with models of prefrontal organisation, these tracts may be responsible for conveying sensory information to prefrontal streams involved in “exploring” new task sets (SLF I and DMPFC), or “exploiting” existing task sets (SLF III

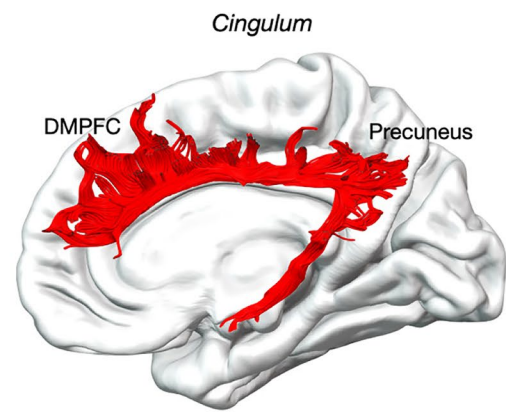
and VLPFC; see Domenech & Koechlin, 2015). Within the prefrontal cortex, regions containing projections of the SLF I (DM or LPFC) and SLF III (VLPFC) are also directly connected within the frontal lobe through the frontal aslant tract (FAT: see also Fig. 3 and Table 1), which runs perpendicular to the SLF branches (Catani et al., 2012; Lawes et al., 2008; Vergani et al., 2014). The FAT connects regions in the cingulo-opercular network (Fig. 2). In the left hemisphere, this tract is associated with more “executive” aspects of linguistic function (for review see Dick et al., 2019). Recently, intraoperative work has demonstrated that

the left FAT is linked to the production of sentences rather than single words, indicating a key role in planning, timing, and coordination of speech (Chernoff et al., 2019; Dragoy et al., 2020). A role in planning is supported by findings in healthy adults that linked interindividual variation in macrostructural features of this tract with lexical decision making (Vallesi & Babcock, 2020). In the right hemisphere, stimulation of this tract causes disruption during the Stroop test (Puglisi et al., 2019). This tract is also regarded to be a core pathway involved in the inhibition of task-irrelevant task sets (Aron et al., 2007). In developing children, a shift

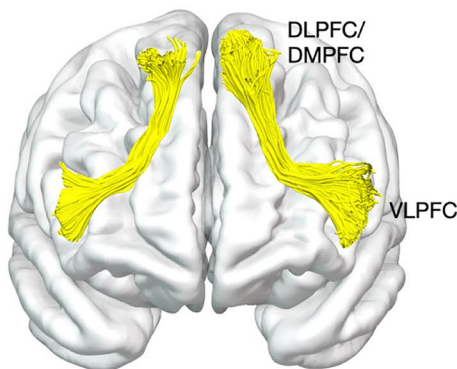
Superior longitudinal fasciculus (SLF) I, II and III



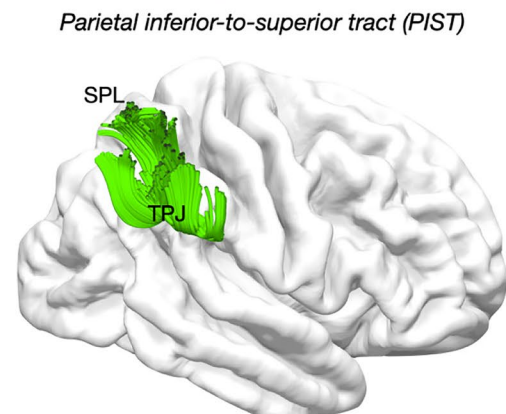
Anterior-posterior connections



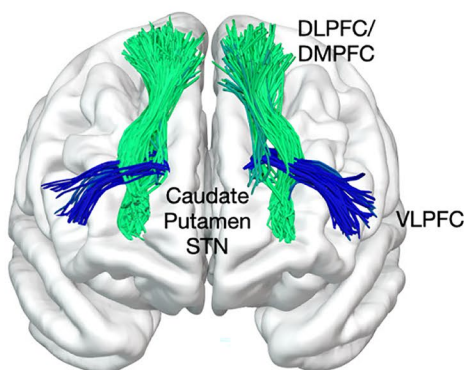
Frontal aslant tract (FAT)



Medial-lateral connections



Fronto-striatal tracts



Subcortical connections

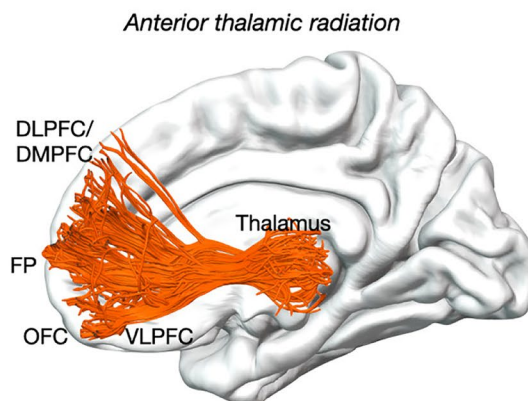


Fig. 3 Major white matter pathways recruited during attention, working memory and executive control processes. Acronyms are defined in Fig. 1

Table 1 White matter tracts associated with attention, working memory and executive functions tasks

White matter pathway	Cortical regions connected
Superior longitudinal fasciculus I (Thiebaut de Schotten et al., 2011)	DLPFC, DMPFC, DLPMC, SMA (superior frontal gyrus) and superior parietal cortex (bilateral)
Superior longitudinal fasciculus II (Thiebaut de Schotten et al., 2011)	DLPFC, DLPMC (middle frontal gyrus) and posterior TPJ (angular gyrus, bilateral)
Superior longitudinal fasciculus III (Thiebaut de Schotten et al., 2011)	VLPFC, VLPMC (ventral precentral gyrus and posterior inferior frontal gyrus) and anterior TPJ (supramarginal gyrus, right-lateralised)
Cingulum (Catani & Thiebaut de Schotten, 2008)	DMPFC, SMA, precuneus, superior parietal cortex and medial temporal lobe (bilateral)
Frontal aslant tract (Catani et al., 2012)	Anterior: DLPFC/DMPFC and VLPFC; posterior: DLPMC/SMA and VLPMC (left-lateralised)
Parietal inferior-superior tract (Catani et al., 2017)	Superior parietal cortex and TPJ (unknown lateralisation)
Anterior thalamic radiation (Rojkova et al., 2016)	Orbitofrontal cortex, frontal pole and thalamus (bilateral)
Fronto-striatal connections (Rojkova et al., 2016)	DLPFC, VLPFC and caudate, putamen and subthalamic nucleus (bilateral)

acronyms refer to cortical labelling in Fig. 1

toward left-lateralisation^G of the FAT is associated with attention and executive function deficits (Garic et al., 2019). Interindividual differences in this tract are also associated with different kinematic components of visually guided hand movements (Budisavljevic et al., 2017) and damage to this tract causes deficits in executing movement sequences (Serra et al., 2017). This suggests the FAT plays an amodal role in the planning and sequencing of actions. Studies of healthy ageing have demonstrated that changes in working memory ability are linked to FAT microstructure (Rizio & Diaz, 2016), which may specifically be linked to the more anterior portion connecting medial and lateral prefrontal regions (Fig. 2, Varriano et al., 2018, 2020).

Furthermore, connections of prefrontal regions with the basal ganglia are well-documented as being involved in executive control processes, particularly in the right hemisphere (Aron et al., 2007). The subthalamic nucleus (STN, Fig. 1), which is a key target for DBS^G due to its role in motor coordination, is also reciprocally connected with the prefrontal cortex and is particularly relevant in visual attention and executive control processes (Schmalbach et al., 2014; Zaghoul et al., 2012). Stimulation of the ventral STN, connected with prefrontal regions, decreases the number of correct responses and increases the number of false alarms during a Go/No-Go task, indicating its relevance for inhibitory control processes. The dorsal striatum, particularly the caudate nucleus, is also relevant in inhibitory control. In the left hemisphere, caudate stimulation causes perseveration errors during language tasks (Robles et al., 2005) and difficulties in switching between languages in bilingual patients (Wang et al., 2013). In the right hemisphere, stimulation of fibres connecting the VLPFC and the caudate causes deficits when patients perform the Stroop test intraoperatively

(Puglisi et al., 2019). Recent models have posited that the basal ganglia act as a gating system for executive processes. When new control demands are required, cortico-striatal circuits modulate the inputs to working memory (Chatham et al., 2014; Frank et al., 2001). This framing would support the broad range of executive deficits that can be induced by stimulation of different regions of the basal ganglia.

Subcortical DES has been used in patients with right frontal gliomas to show that stimulation of the SLF I results in errors on a spatial n-back task (Kinoshita et al., 2016). Using both digit span^B and visual n-back^B intraoperative tasks has also shown that the DLPFC, particularly the right superior frontal gyrus (SFG), supports both verbal and spatial working memory processes (Motomura et al., 2019).

Observations from intraoperative mapping also support and expand upon the previously proposed idea that the SLF III is involved in the phonological loop of working memory (Baddeley, 2007). When subcortical stimulation is applied to the left SLF III, errors are induced during an intraoperative digit span^B task (Papagno et al., 2017).

The medio-dorsal nucleus and other anterior thalamic nuclei are reciprocally connected with prefrontal regions through the anterior thalamic radiations (Fig. 3 and Table 1) and lesions cause similar dysexecutive deficits to those observed following prefrontal lesions (De Witte et al., 2011; Van der Werf et al., 2003). The use of a delayed-response task to test working memory in non-human primates has shown changes in firing rates in prefrontal and anterior thalamic regions (Fuster & Alexander, 1971). When deep brain stimulation (DBS) is applied for the treatment of refractory epilepsy in humans, the DBS lead targeting the anterior nucleus of the thalamus (superior and anterior to the medio-dorsal nucleus), also affects the medio-dorsal nucleus. Using an integrated test for working

memory, emotion, integration, and shifting (the Executive RT test^B), one study has demonstrated that stimulation of anterior thalamic nuclei causes disruption to attentional processes for emotional stimuli (Sun et al., 2015), while stimulation of the medio-dorsal nucleus causes specific impairment to working memory processes (Peräkylä et al., 2017). This suggests that the dense innervation of prefrontal regions through thalamic fibres are key for attention and executive control in humans, supporting the considerable body of evidence showing this in non-human species (for reviews see Mitchell, 2015; Pergola et al., 2018).

In humans, electrical stimulation of the left pulvinar and ventrolateral thalamic nuclei causes verbal working memory deficits (Ojemann & Fedio, 1968; Ojemann et al., 1971). Hence, thalamic connections enable the rapid coordination of different cortical regions that subtend executive control (Halassa & Kastner, 2017).

Integrating tractography with other functional methods demonstrates that neural processing of attention, working memory and executive control share common highways of information flow through white matter pathways. This further indicates these processes are the result of the integrated function of networks recruiting widespread brain regions, rather than through specialised function of localised areas.

Invasive Brain Recordings

Functional and structural magnetic resonance imaging lack the ability to characterize fine-grained temporal dynamics of cognitive functions, while other non-invasive brain imaging methods, such as electroencephalography and magnetoencephalography, allow the investigation of the spatiotemporal activation, oscillation, and connectivity of large neuronal populations. These methods have been extensively used to study the dynamics of attention (Bae & Luck, 2018; Favaretto et al., 2021; Spagna et al., 2022), working memory (e.g., Zhang et al. 2020; Onton et al., 2005), executive function (Mizuhara & Yamaguchi, 2007), and their functional overlap has already been reviewed elsewhere (Viviani & Vallesi, 2021; Luck et al., 2010; Menon & D'Esposito, 2022; Gratton et al., 2018). Therefore, we will focus here on invasive brain recordings that provide even greater spatiotemporal resolution for the investigation of cognitive functions, with a trade-off for whole-brain coverage.

The independence and interactions of attention, working memory, and executive control processes have been examined using invasive intracranial electroencephalography (see iEEG^G) recordings. In humans, these recordings are performed in a unique population of patients with intractable epilepsy as a part of their medical treatment, and allow investigation of the direct neural correlates of

cognitive processes with unprecedented spatiotemporal resolution (Helfrich & Knight, 2019; Mukamel & Fried, 2012; Parvizi & Kastner, 2018). Over the last decade, the high-resolution exploration of such neural information over different scales (from single units to large populations of neurons) has led to the emergence of novel ways for conceptualizing and physiologically explaining cognitive processes and their interactions. One important finding is that cognition and its underlying brain activity are not constant over time, but rather exhibit intrinsic temporal profiles that rhythmically wax and wane (Helfrich & Knight, 2019). The synchrony of these neural oscillations^G across different frequency bands^G has been proposed to be a potential mechanism for information integration over local and long-distance neural populations (Helfrich & Knight, 2019; Jensen et al., 2007; Varela et al., 2001).

For example, attention has been conceptualized as a rhythmic process in which cueing a spatial location increases the sampling frequency of the brain from that location, and by aligning the oscillatory optimal phase, the processing of the cued-location information could be enhanced through endogenous entrainment or phase resetting (Helfrich & Knight, 2019; VanRullen, 2016). This hypothesis has been supported by non-invasive human electrophysiological findings (Fiebelkorn et al., 2013; Landau & Fries, 2012; Landau et al., 2015; Song et al., 2014), demonstrating that visual perception at a cued location is sampled at approximately 8–12 Hz (alpha band), while unattended spatial positions are only monitored at a slower rhythm (4 Hz, delta or theta band). Notably, high frequency activity (HFA, also known as broadband gamma, 50–150 Hz), a marker for multi-unit neural activity (Ray & Maunsell, 2011), has been associated with various cognitive processes (Helfrich & Knight, 2019; Jensen et al., 2007). HFA synchrony has been proposed as a putative physiological mechanism for directed attention and working memory (Jensen et al., 2007). According to the principle of binding by neuronal synchronisation, during attention processes, groups of neurons encoding an attended stimulus will fire more synchronously than groups of neurons encoding an unattended stimulus, and therefore will provide a stronger drive to downstream targets compared with less-synchronized neurons (Jensen et al., 2007). Conversely, during working memory operations, a set of neurons encoding a specific sample stimulus will fire in synchrony during stimulus presentation and will maintain synchronous oscillatory firing during the retention interval, thus providing a mechanism for working memory maintenance (Jensen et al., 2007; Lundqvist et al., 2018; Miller et al., 2018). Other frequency bands^G were also found to be implicated in these processes (Fiebelkorn & Kastner, 2019; Helfrich & Knight, 2019). In particular, theta-band activity was modulated by both

working memory and attention, and its phase was found to be modulating broadband gamma-frequency power (Canolty et al., 2006).

In agreement with fMRI^G results emphasizing the role of fronto-parietal networks in attention functions, evidence from iEEG^G recordings substantiates the involvement of rhythmic activity in fronto-parietal but also in sensory regions during various attention processes (Helfrich & Knight, 2016, 2019; Jensen et al., 2007). For example, Ossandón et al. (2012) reported that pop-out visual search was associated with sustained increases in activity of the DAN^G-HFA throughout the duration of the search both in efficient and inefficient search conditions, supporting the hypothesis that active visual search systematically involves executive attention resources, regardless of target saliency. However, attentional modulation of HFA might differ in its time-course and direction of modulation across the brain. For example, in the lateral occipital cortex (Fig. 1), attention amplified HFA during preparatory period, while in the fusiform gyrus it caused a decrease during stimulus processing (Tallon-Baudry et al., 2005). Amplification of HFA by attention was also found outside sensory cortices, for example, in the dorsolateral premotor cortex (DLPFC, Fig. 1) of participants shifting their visual attention towards the location of an upcoming target (Brovelli et al., 2005). The involvement of theta-HFA cross-frequency oscillatory activity in sustained attention was recently explored in a study demonstrating that rhythmic behavioural fluctuations at a cued location are the direct result of HFA that is modulated by ongoing theta-band oscillations^G over fronto-parietal networks (Helfrich et al., 2018). Specifically, participants' performance in two different sustained attention tasks showed that theta-band fluctuations could be predicted by the phase of ongoing theta-band frontoparietal activity during the cue-target-interval. Two recent studies established the spatiotemporal dynamics of HFA power during the orienting and reorienting of exogenous attention (Liu et al., 2022; Seidel Malkinson et al., 2021). Using a data-driven approach to reveal clusters of iEEG contacts that share similar HFA activity patterns across all experimental conditions simultaneously based on their temporal trajectory dynamics, Seidel Malkinson et al. (2021) found that exogenous attention emerged over a large-scale spatiotemporal gradient of neural clusters embedded in the cortical gradient topography along which perceptual and response signals converge, and that information integration along this cluster gradient occurred over longer and longer timescales. Using the same unsupervised clustering method, Liu et al. (2022) explored the interplay between exogenous attention and conscious perception of near-threshold visual stimuli and revealed that attention-consciousness interactions were associated with distinct HFA activity dynamics across frontoparietal networks in a hemisphere specific pattern. While the conscious report of

cued targets was associated with sustained *right-hemisphere* fronto-temporal activity and with late accumulation in bilateral dorso-prefrontal and *right-hemisphere* orbitofrontal cortex, conscious report of uncued targets, requiring attentional reorientation, was associated with early, sustained activity within a *right-hemisphere* reorienting network. Task modelling with recurrent neural networks established the causal contribution of these networks to conscious perception, and elucidated their inter-network interactions. Based on such findings, Fiebelkorn & Kastner (2019) have recently put forth a rhythmic theory of attention, stressing the importance of theta rhythms from the frontoparietal attention networks, which are positioned at the nexus of sensory and motor functions, in temporally organizing sensory and motor functions and creating alternating periods of either enhanced or diminished perceptual sensitivity. While theta-dependent periods of enhanced perceptual sensitivity reflect attention-related sensory sampling, theta-dependent periods of reduced perceptual sensitivity provide opportunities to reorient attention, reassessing periodically the current importance of the attended location (Fiebelkorn & Kastner, 2019). Based on evidence from monkey electrophysiology (Fiebelkorn et al., 2018), Fiebelkorn & Kastner (2019) proposed that enhanced perceptual sensitivity periods are characterized by theta-dependent increase in FEF-dominated beta-band activity (15–35 Hz) and lateral intraparietal (LIP)-dominated gamma-band activity (> 35 Hz), repeatedly associated with sensory enhancement and oculomotor suppression, respectively. Hence, this attentional enhancement period may be characterized by the enhancement of sensory processing and the suppression of attentional shifts (either covert^G or overt^G). On the other hand, reduced perceptual sensitivity periods were found to be associated with LIP-specific alpha-band activity, repeatedly linked to the attenuation of sensory processing. This reduced processing at the attended location along with the temporary release from beta-related motor suppression could periodically increase the likelihood of attentional reorientation (Fiebelkorn & Kastner, 2019). Rhythmic neural activity was also implicated in executive control. Smith et al. (2019) examined the responses of single neurons and local field potentials in human neurosurgical patients in the dorsal ACC (dACC) and DLPFC during the performance of a conflict task, and found a widespread effect of conflict on spike-phase coupling in the dACC and on driving spike-field coherence in the DLPFC, supporting the hypothesis that a cross-regional rhythmic neuronal coordination is intrinsic to executive control in response to conflict. Gamma-oscillations^G were also associated with executive control. Using ECoG (or iEEG^G), Koga et al. (2011) found increased gamma-power over the DLPFC, DLPFC, and SMA in the Stroop test. This study has provided direct evidence that executive control in the context of the Stroop test depends on oscillatory broadband gamma

activity over the dorsolateral premotor and prefrontal areas. The supplementary motor area was also shown to have a leading role in error detection, using iEEG^G recordings during the performance of the Simon task^B (Bonini et al., 2014). Theta oscillations^G were also suggested to be pivotal to executive control. Considering evidence from non-invasive EEG, Cavanagh and Frank (2014) suggested that theta oscillations^G over the mid-frontal cortex may reflect a common computation used for realizing the need for executive control. Moreover, theta-oscillations^G may serve to communicate this need and subsequently implement such control across diverse brain regions.

Human iEEG^G recordings show that theta (Meltzer et al., 2008; Raghavachari et al., 2001, 2006; Zhang et al., 2018), alpha (Meltzer et al., 2008) and broadband gamma (Howard et al., 2003; Jensen et al., 2007; Mainy et al., 2007; Meltzer et al., 2008) band oscillations^G spanning across frontal, parietal, and sensory cortices are also associated with working memory (See Roux & Uhlhaas, 2014 for a review). Working memory related oscillations^G were localized to the fronto-parietal networks, however their role seems to be modulatory (e.g., increasing or decreasing information gain according to task relevance), suggesting that working memory operates as a distributed network involving also sensory areas in memory maintenance (D'Esposito & Postle, 2015; Lara & Wallis, 2015). For instance, gamma oscillations^G in frontal, temporal, parietal, and occipito-temporal cortices were shown to increase approximately linearly with memory load in implanted epileptic patients performing a delayed match-to-sample task (Howard et al., 2003). In another study, Zhang et al. (2018) used ECoG to explore retrieval and maintenance of faces in working memory and found that while activity in the medial temporal lobe reflected active maintenance of the studied face set in the decision stage, the frontal cortex guided the decision by means of theta coupling with the medial temporal lobe. Recently, Miller et al. (2018) proposed a revision of the classical model of working memory, suggesting that the top-down control of working memory content arises from the interaction between oscillatory patterns in different frequency bands (i.e., gamma and alpha or beta). Specifically, by reducing the power of alpha or beta activity sourcing from deep cortical layers, the connection between excitatory (pyramidal) and inhibitory (interneurons) allows gamma oscillation bursts from the superficial cortical layers, resulting in enhanced working memory performance. Moreover, studies of various sensory modalities have also suggested that the site of oscillatory activity underlying working memory depends on the type of information that must be retained. iEEG^G recordings during maintenance of olfactory identity information revealed sustained gamma-frequency oscillations^G in the amygdala (Jung et al., 2006), while auditory working memory tasks elicited oscillatory activity in posterior temporo-parietal (Lutzenberger

et al., 2002) and left inferior frontal and anterior temporal cortices (Kaiser et al., 2003). iEEG^G findings regarding the distributed nature of working memory dovetail the results of a recent study using non-invasive EEG to explore oscillatory activity in individuals with PFC lesions while they performed a working memory task (Johnson et al., 2017). In this study, the authors found that compared to control participants, individuals with PFC lesions showed lower task accuracy (yet above chance), weaker PFC low-theta activity, and reduced PFC to parieto-occipital regions connectivity, yet had preserved alpha–beta oscillations^G over parieto-occipital regions. This intact dissociable alpha–beta parieto-occipital system might be the neural basis of the patients' reminiscent working memory performance (Johnson et al., 2017). On a similar note, Roux and Uhlhaas (2014) proposed that different frequency bands^G serve distinct roles in WM across the brain. For instance, gamma-band oscillations^G are generically involved in the maintenance of working memory information, alpha-band activity reflects the active inhibition^G of task-irrelevant information, and theta-band oscillations underpin the organization of sequentially ordered working memory items (Roux & Uhlhaas, 2014). Taken together, gathered evidence suggests that oscillatory activity is associated with attention, working memory, and executive control, especially in broadband gamma and theta frequencies, as well as in the interactions between these bands. As discussed in the previous sections of this manuscript, the fronto-parietal cortex is implicated in these oscillations^G across the three cognitive domains (Fiebelkorn & Kastner, 2019; Helfrich et al., 2018; Helfrich & Knight, 2019; Howard et al., 2003; Koga et al., 2011; Miller et al., 2018; Smith et al., 2019; Zhang et al., 2018). In general, synchronization in broadband gamma frequency might serve as a binding mechanism driving the enhancement of stimulus representation, and theta oscillations^G may serve as a modulating mechanism, coordinating broadband gamma activity across distant regions. Indeed, Fiebelkorn & Kastner (2019) suggested that theta oscillations might not be an attention specific phenomenon, but could be a more general mechanism providing functional flexibility in large-scale network-based processes. For example, in the case of working memory, the brain might rhythmically cycle through stronger neural representations of each to-be-remembered item, rather than fully process the items simultaneously. Hence, lower frequency oscillations^G might have a general function in organizing and disassociating functionally distinct, large-scale networks that have overlapping hubs^G (Fiebelkorn & Kastner, 2019). These oscillations^G are a potential candidate mechanism by which emergent processes such as “executive control” may be biophysically realized (Cavanagh & Frank, 2014). A striking example of the power of invasive brain recordings in dissecting the overlap and distinction of executive control functions is a recent study showing that neurons

in the prefrontal cortices of monkeys that focus attention on sensory stimuli are the very same ones that focus on an item in working memory (Panichello & Buschman, 2021). By contrast, in the parietal and visual cortex, both attention and working memory were represented independently. Furthermore, when an item was selected, its memory representation in the prefrontal cortex dynamically rotated, such that the representation was either hidden away in working memory or used to guide behaviour, thus allowing the cognitive control of behaviour (Panichello & Buschman, 2021). Future iEEG research in humans is needed to extend these findings to the human brain and to explore for example how the rotation of neural representation interacts with oscillatory mechanism of executive control.

Stimulation Approaches

Non-invasive Brain Stimulation

Non-invasive brain stimulation technologies enable transient inactivation of focal cortical regions (producing “virtual lesions”), while recording behavioural consequences of such actions (see Valero-Cabré et al., 2017). By causally manipulating neural activity, methods such as transcranial magnetic stimulation (TMS), transcranial direct current stimulation (tDCS), transcranial alternating current stimulation (tACS), intermittent or continuous theta burst stimulation (cTBS) are particularly well suited to investigate brain-behaviour relationships. These methods have also been used in the investigation of attention, working memory, and cognitive control.

Non-invasive stimulation studies have confirmed the role of the fronto-parietal networks in attention. Specifically, TMS perturbation of the FEF, known especially for its role in the control of eye movements (see Pierrot-Deseilligny et al., 2004), have emphasized the involvement of this region in the top-down control of attention (see Grosbras & Paus, 2003; Hung et al., 2011; Silvanto et al., 2006) and sustained attention (Esterman et al., 2015), thus confirming observations from fMRI^G studies (Corbetta & Shulman, 2002; Corbetta et al., 2008; Fan et al., 2005; Lawrence et al., 2003; Xuan et al., 2016). Interestingly, non-invasive stimulation studies uncovered an important lateralization^G of functions of the FEF (see Murd et al., 2020), with the right hemisphere FEF presenting a more general role in top-down control of attention and contributions to fluctuations of sustained performance. Specifically, inhibitory repetitive TMS (rTMS) on the right FEF worsened sustained attention, provoking lower accuracy and increased RT variability, while offline inhibitory rTMS on the left FEF did not affect accuracy and variability (Esterman et al., 2015). In the same vein, TMS over the right FEF reduced attentional control by increasing distractor interference on both sides of space, whereas TMS

over the left FEF had no significant effect on performance (Hung et al., 2011). Recently, Lega et al. (2019) evaluated the causal role of key regions of the DAN^G in resisting attentional capture by a salient distractor and emphasized the role of the right FEF in distractor-filtering. Other TMS studies (e.g., Chanes et al., 2012, 2013; Chica et al., 2014; Quentin et al., 2015, 2016) highlighted the ability of the FEF to enhance the conscious detection of visual stimuli and contribute to attentional orienting (see Vernet et al., 2014 for a review). Not only the FEF, but also the intraparietal sulcus in the DAN^G, has been manipulated in TMS studies to show a causal role for this region in both endogenous^G and exogenous^G attentional orienting (Chica et al., 2011). Further, these authors also reported that the TPJ (part of the VAN) was recruited for exogenous^G attentional orienting, but only when stimuli were behaviourally relevant for the task at hand (Chica et al., 2011; Kincade et al., 2005 for a confirmatory fMRI study, and Stengel et al., 2022 for a recent review). In order to improve attentional control, Esterman et al. (2017) manipulated functional connectivity within the DAN^G using intermittent theta-burst TMS administered to the midline cerebellar node of this network, localized using individual resting-state fMRI^G scans. Results showed a reduction of attentional lapses, suggesting that temporarily increasing connectivity in the DAN^G via network-targeted cerebellar stimulation can enhance sustained attention, particularly in individuals with lower baseline performance. TMS stimulation in both the DAN^G and *default mode network* (DMN^G) improved *attentional blink* performance (the decrease in detection rates of a target stimulus if presented shortly after [i.e., below 200 ms] another imperative stimulus), suggesting a link between these networks and attention. Other TMS studies have assessed the role of the DAN in the attentional control of temporal preparation (Vallesi et al., 2007). By stimulating the DLPFC, these authors noticed a significantly reduced foreperiod effect after TMS over the right DLPFC but not over the left DLPFC, suggesting a lateralized role of this region in monitoring temporal processes.

Interestingly, TMS studies emphasize the supramodal (independent of the stimulus modality, whether verbal or spatial) role of the left DLPFC in working memory and executive control (see Mottaghy, 2006), suggesting the existence of a supramodal central executive within the left DLPFC, a claim that is in line with behavioural (Spagna et al., 2015, 2018) and fMRI^G (Duncan, 2013) results.

As in the fMRI studies, working memory systems have been investigated with n-back^B tasks in non-invasive stimulation studies. In agreement with fMRI^G findings showing activations of the fronto-parietal networks (PMC, DLPFC, VLPFC, DMPFC, frontal poles, and TPJ) during this task (see Owen et al., 2005 for a review), the meta-analysis of Brunoni and Vanderhasselt (2014) reported that non-invasive stimulation on the DLPFC was associated with faster

RTs, a higher percentage of correct responses and a lower percentage of error responses in this task. A comparison between stimulation methods showed that rTMS on the DLPFC generally improved the working memory measures, whereas the tDCS only improved RTs in the n-back^B task. More recently, Lynch et al. (2019) used continuous theta-burst TMS to inhibit hub^G and non-hub areas (identified using resting state fMRI) of the working memory system in the middle frontal gyrus. These authors observed that working memory performance was more disrupted for hubs^G than for non-hub areas, despite anatomically adjacent targets, suggesting anatomical specialization for the working memory system.

Further, in order to causally investigate a fMRI^G-based model of executive control, Nee and D'Esposito (2016, 2017) used continuous theta-burst TMS over three regions of the lateral PFC (LPFC), including caudal DLPFC (superior frontal sulcus, SFS), mid LPFC (VLPFC), and rostral LPFC (lateral frontal pole). These authors reported that cTBS to caudal DLPFC disrupted attention to stimulus feature (feature control) and the ability to balance ongoing and future demands (temporal control). Continuous theta-burst TMS to the VLPFC disrupted responses to stimulus features based on contextual demands (i.e., contextual control), and continuous theta-burst TMS to the frontal pole disrupted again the temporal control, suggesting a modulation from caudal LPFC to mid-rostral LPFC. A recent review of studies that have used non-invasive brain stimulation to assess executive control in the stimulus–response compatibility tasks (Simon, Flanker, and Stroop paradigms: see Olk et al., 2015) reported different fronto-parietal contributions. The FEF encodes the spatial attributes of stimuli, the angular gyrus mediates the allocation of spatial attention and orienting, while the dorsal medial frontal cortex, supramarginal gyrus, and pre-SMA are more related to response-related aspects of the conflict (i.e., enhancement of signals related to correct movements, transformation of spatial information action codes, and resolution of response selection conflicts).

Executive control was investigated in other studies, with the verbal fluency task. Smirni et al. (2017) reported that inhibition over the left lateral frontal cortex significantly worsened letter-cue fluency^B performance when compared to the sham condition, whereas rTMS inhibition over the right lateral frontal cortex significantly improved letter-cue fluency performance when compared to the sham condition, likely because of the increased excitability (i.e., disinhibition) of the contralateral unstimulated left lateral frontal cortex (see Valero-Cabré et al., 2020 for a review of the mechanisms responsible of such effects).

Taken together, all these data suggest common underpinnings of attention, working memory, and executive control networks involving interactions of structures within and between DAN and VAN networks, as well as their

interactions with other networks, as shown by the cerebellar stimulation study (Esterman et al., 2017). Experimental perturbation-based approaches using non-invasive neurostimulation are fundamental in the causal exploration of all these functions and well suited to identify interesting hypotheses, but these hypotheses need to be confirmed by convergent evidence from other methods such as lesion inference and invasive cortical stimulation that will be developed further.

Cortical Invasive Brain Stimulation

While neuroimaging and electrophysiological methods are all invaluable to the study of brain-behaviour relationships, these methods are limited in their ability to demonstrate the necessity of any particular neuroanatomical region or structure to the functional network in question. While lesion studies can demonstrate this necessity, lesions are commonly non-focal to specific functional regions and may be limited to the vascular distribution in the case of stroke (Arnoux et al., 2018; Godefroy et al., 1998). The use of direct electrocortical stimulation^G (DES) in neurosurgery to induce transient virtual lesions in vivo is therefore a useful tool to demonstrate the functional eloquence of highly focal cortical and subcortical regions (Duffau, 2011; Herbet & Duffau, 2019; Szelényi et al., 2010). DES is most frequently applied in the context of awake craniotomy for surgical resection of primary brain tumours, or gliomas, in or near canonical sensorimotor or language circuitry. From a neurosurgical perspective, much of the right hemisphere has long been considered ‘silent’, because relative to the more impairing disabilities that can occur following left hemisphere resection of dominant language and motor regions, right hemisphere resections outside of eloquent cortex are typically associated with more subtle neurological deficits. Consequently, right hemisphere functionality has been understudied in comparison to the left hemisphere (Voets & Plaha, 2019), and the literature on the use of DES to probe networks underlying higher level cognition is still limited.

Brain mapping to identify eloquent regions for visual attention is commonly performed using a simple test of visuospatial attention, namely, the line bisection test^B (Bartolomeo et al., 2007; Thiebaut de Schotten et al., 2005), which has demonstrated that both cortical and subcortical regions play an important role. In particular, deviations on the line bisection^B occur intraoperatively when DES is applied to the VLPFC, TPJ, and the posterior superior and middle temporal gyrus in the right hemisphere (Roux et al., 2011; Vallar et al., 2014).

While spatial working memory has not been studied on the cortical level in the awake setting, there has been some cortical investigation to localise verbal working memory functions using DES in the left hemisphere. One study used the digit span^B task and identified the type of error (item

processing or serial order) that occurred following cortical stimulation of the posterior inferior frontal gyrus and the supramarginal gyrus (Papagno et al., 2017). The results showed selective disruption of digit span^B in different cortical sectors. Stimulation of the posterior inferior frontal gyrus primarily caused item processing errors, while stimulation of the supramarginal gyrus caused interference with order processing. The authors hence theorised that the posterior inferior frontal gyrus may act as a phonological output buffer while the supramarginal gyrus is part of the phonological short-term store, following Baddeley's model of working memory (Baddeley, 1992).

Similarly to the working memory literature, very few studies have investigated the neural correlates of executive control in the awake surgical setting, and have primarily used the Stroop test^B. One study showed that deficits on this test occur following targeted stimulation of the ACC, bilaterally (Wager et al., 2013). A second study investigated the response to the Stroop test with right VLPFC stimulation only, and could not identify any interference when applying cortical stimulation. Rather, this effect was specific to subcortical stimulation in this region, particularly beneath the inferior frontal sulcus (Puglisi et al., 2019). Stimulation of the left DLPFC caused errors on a cross-modal discrimination task, specifically on making visual-verbal congruency judgements of pictures, providing causal support for the involvement of this region in decision making and executive control (Plaza et al., 2008).

The observations of the small but growing body of literature on intraoperative brain mapping of attention and executive control functions support the shift that has taken place in the conceptualization of brain-behaviour relationships from a strict localizationist perspective toward a network-based or hodological approach (Catani, 2007). In the previous example, the discrete effects on the digit span^B task following cortical stimulation in different brain regions suggest that the task requires both components, hence the connection between these regions is crucial. As awake surgery for brain tumours necessarily requires subcortical investigation, this has enabled causal investigation of the white matter connections supporting cognitive processes.

Summary and Future Directions

Although outstanding questions regarding the brain mechanisms supporting complex behaviour remain (see Box 2), growing evidence indicates that attention, working memory, and executive control do share functional and structural neural correlates. The goal of the present review was to provide a contemporary view integrating

updated cognitive models, multi-modal imaging, and brain stimulation toward an attempt to synthesize the past 50 years of research on this topic. This review highlights two main points: (a) the existence of meta-networks supporting executive control, and (b) the existence of a lower dimensional space for functions supporting goal-directed behaviour.

Box 2 Outstanding questions.

By the end of this review, we agreed that some key questions still stand, and have decided to provide them below, in an effort to begin fostering the collaborative environment required to fully achieve the multimodal approach detailed in the summary section:

1. Can the recent advancement in collecting “*big data*” enhance the clinical implications of the study of executive control?
2. How can we reconcile current literature with the evidence—although anecdotal—of patients with frontal and prefrontal lesions that do not show executive control deficits?
3. As with many other scientific fields, the neuropsychological research endeavour has been extensively dominated by the *white male* figure. How would increasing diversity and inclusion in the field of executive control change our current understanding of the control functions and the underlying neural substrates?
4. As the interest in *real-world* neuroscience grows (i.e., research on cognitive functions conducted in ecologically valid environments such as in virtual reality), how valid will our knowledge of higher cognitive function be outside of current laboratory settings?
5. To what extent will applications of network neuroscience complement lesion and functional neuroimaging studies in order to better define interactions between attention, working memory and executive control? What can neuropsychology of executive control learn from the network neuroscience field?
6. How to reconcile classical neuropsychological evidence and functional neuroimaging results showing the critical role of ACC in WCST performance with intact—if not improved—WCST performance following surgical intervention in ACC and DLPFC (e.g., in the case of cingulotomy patients (see Jung et al., 2006)?
7. The fundamental disconnect between clinical neuropsychology and cognitive neurosciences arises from the lack of well-validated normative data for research-based tasks to be used in clinical populations. Bridging this gap is essential to the future of neuropsychology. What will it take to be able to do so?

A Multi-method Study of Executive Control Reveals the Existence of Meta-networks

The literature reviewed here suggests that a multi-modal approach is critical to see a full picture of structural and functional data reported in the attention, working memory, and executive control fields, and emphasizes the need to move away from strict localization toward hodotopical approaches. Functional task-based MRI and resting state approaches initially contributed to this shift (Pessoa, 2014; Uddin et al., 2019) with the identification of roles of the fronto-parietal and cingulo-opercular networks in attention, working memory, and executive control. Diffusion tractography combined with functional invasive and non-invasive

methods has helped to identify the anatomical substrates of functional connectivity between cortical and subcortical hubs involved in executive control. Invasive electrophysiological recordings have helped understanding of how oscillatory activity in these hubs drives stimulus representation and coordinates oscillatory activity across distant regions, resulting in flexible network states. Integrating functional neuroimaging literature, invasive and non-invasive stimulation research, and studies of patients with focal lesions within the fronto-parietal and cingulo-opercular networks adds causal evidence for the involvement of each brain region in these functions (Vaidya et al., 2019), uncovering the neural architecture supporting flexible goal-directed behaviour.

Executive control studies are also benefiting from network neuroscience as a means to understand the brain-behaviour relationship, by adopting a connectional systems perspective and considering behaviour as the result of interactions between hierarchically organized brain networks (Bassett et al., 2018; Bullmore & Sporns, 2009; Pessoa, 2014; Uddin et al., 2019). Applying this new conceptual and methodological framework to executive control also helps the characterisation of its anatomical correlates in a revisited form, and provides new insights into the reorganisation of networks in neurological diseases (Fox, 2018; see Glasser et al., 2016; Jha et al., 2020 for examples of new approaches to be extended to the study of the executive control; and Toba et al., 2020a for further conceptual discussion). Expanding on a recently proposed taxonomy (Uddin et al., 2019) that considers anatomical brain networks as a basis for describing functional networks, “core” networks of executive control can be established. On the basis of the literature reviewed here, cross-modal convergence supports evidence that core networks may include the (a) lateral fronto-parietal network, (b) the dorsal fronto-parietal network, (c) the medial fronto-parietal network, and (d) the midcingulo-insular network. These large-scale networks flexibly assemble and interact in order to provide the dynamic and hierarchical nature of executive control. Importantly, there appears to be incomplete correspondence between these large-scale networks identified using resting-state fMRI and executive control networks reported in task-based or activation studies (Cole et al., 2014; Bolt et al., 2017; Uddin et al., 2019). However, it has been suggested that these resting-state networks broadly distributed across the cortex could represent the full functional repertoire, from which executive control tasks may engage only sub-networks or circumscribed subsets of regions (Yeo et al., 2015; Uddin et al., 2019). Furthermore, these sub-networks do not act in isolation. Rather, their interactions, and consequently a mixture of sub-functions they subtend, are essential for an efficient deployment of executive control.

Interactions between and within networks result in a perpetual dynamic reconfiguration of networks during

complex cognitive demanding tasks, involving transitory, task-induced changes in integration between brain regions encompassing specialized functional systems (Cocchi et al., 2013). This perspective has led to the concept of “meta-network” (i.e., networks of networks, see Herbet & Duffau, 2020), which refers to the ability of brain networks to flexibly reconfigure as a function of the ongoing behavioural task, and is characterized by specific patterns of large-scale between-systems integration (D’Esposito, 2007). This meta-network has been related to functions such as working memory (Shine et al., 2016), cognitive flexibility (Douw et al., 2016), sustained attention, and cognitive control (Shine & Poldrack, 2018). A meta-network perspective implies the existence of cortical nodes that may be highly connected, which is especially important for cross-network integration (de Pasquale et al., 2016; Herbet & Duffau, 2020). In the executive control network, these nodes may include the DLPFC and the cingulate cortex, the nodes having a central role in coordinating functional interactions between segregated networks (see for instance van den Heuvel et al., 2009 and Kinoshita et al., 2015, for interactions between the lateral and medial fronto-parietal network networks). Applying this perspective to executive control may lead to novel predictions that would move the field forward by focusing on the interactions between sub-functions or sub-networks, and by constructing a model that portrays executive control as the output (i.e., the result) of efficient allocation of control resources and implementation of processes.

Mapping a Low-dimensional Space of Executive Control Anatomy and Function

From a conceptual point of view, reducing the number of executive control sub-functions and reporting commonalities between these sub-functions would enable dimensionality reduction, and enable the current perspective to be reframed based on anatomo-functional data (see Karolis et al., 2019). These changes are being facilitated through the increasing classification and communication of findings across research groups and the sharing of data to allow the study of cognitive functions. These changes should enable the joint study of executive control and brain-based mechanisms to move forward coherently.

The classical approach to investigating associations between brain and behaviour consists of choosing a subjectively defined term (such as inhibition, energization, etc.) as the independent variable of the study, and looking for dependent variables in the brain. However, a critique to this approach has been recently brought forward (see Buzsáki, 2020), centered on evidence that the cognitive terms used in the cognitive neuroscience literature are subjectively defined and rooted in the philosophical field rather than from an objective investigation of the brain (Buzsáki, 2020). Consequently, identifying an association between psychological

mechanisms (in our case, executive control) and brain regions (e.g., prefrontal cortex) under this classical approach may be misleading. Buzsáki (2020) proposes a flipped approach, starting from a distinct brain pattern (i.e., the independent variable), then using anatomo-functional data to search for and define external correlates (i.e., descriptions of behaviour) that could represent the dependent variable (Buzsáki, 2020). This approach could allow the definition of an objective vocabulary of executive functions to use in relation to brain structure and function.

The aim of the present review paper was to provide a synthesis of the literature suggesting the presence of a lower dimensional space in the executive functions domain. Further studies following this line of research should provide direct experimental evidence of this theoretical construct in both neurotypical (adult and developmental) and clinical populations, and suggest newer models of executive functions based on this ground. Of note, recent data-driven studies re-analysing behavioural findings in order to identify best-fitting models of factors underlying executive functions (see Karr et al., 2018 and Eisenberg et al., 2019) refute classical taxonomies of executive functions and propose that lower dimensionality might better account for adults' performances (e.g., Fournier-Vicente et al., 2008), while unidimensional models would be more suited for performances observed in child or adolescent samples (e.g., Brydges et al., 2012). These findings also suggest that individual differences in executive tasks performance may have a lower dimensionality than was typically assumed (Rouder & Haaf, 2019). These data also emphasize the need for new data-driven models in the cognitive science domain based on larger samples, supported by clearer neuropsychological measures, ideally representing orthogonal constructs, which are able to avoid the measurement errors recently emphasized in different tasks (see for instance Hedge et al., 2018; Draheim et al., 2019; Rey-Mermet et al., 2019; Rouder & Haaf, 2019, for the stimulus response compatibility tasks) and possible sampling errors (see for details Karr et al., 2018). As we look to the future, in the new cognitive computational neuroscience framework (Kriegeskorte & Douglas, 2018), we can anticipate that the need for a lower dimensionality of neurocognitive mechanisms in executive functions will require a different, quantitative approach, based on formal generative models able (a) to explain behaviour without the need of subjective defined terms, and (b) to formulate quantitative behavioural and brain association predictions. Existing models embracing this approach have successfully translated behaviour data into components of cognitive processing (see the diffusion decision model for an example, Ratcliff & McKoon, 2008; Ratcliff & Rouder, 1998) and propose to associate intact or lesioned brain regions with cognitive processes or

neurological deficits (see the brain modes for an example, Godefroy et al., 1998; Toba et al., 2020a).

This framework also has implications for investigating the development of executive control, mechanisms of executive dysfunction, and applications to intervention. It stands to reason that developmental or acquired perturbations in executive control network and meta-network integration can account for the heterogeneous presentation of executive dysfunction in clinical populations. Together, this framework suggests a role for strengthening meta-network interactivity as an application for intervention in disorders of executive control (see Tang & Posner, 2014; Cahn & Polich, 2006; Rubia, 2009; Xue et al., 2014 for mindfulness meditation).

In sum, the present review provides evidence supporting cohesion in the executive control field, moving toward dimensionality reduction in the classification of cognitive functions, and the brain regions and connections supporting them. The timeliness of this issue in the context of the developments in the network neuroscience domain will allow closer exploration of large-scale networks and their behavioural expression repertoires in executive control.

Glossary

'Attentional control of action' model Norman and Shallice (1986) proposed the existence of two attentional control processes: (a) the contention scheduling mechanism that selects between conflicting routine actions, and (b) the supervisory attentional system (SAS) that intervenes when the selection of routine actions is not sufficient (as in novel situations). The authors suggested that a deficit of a supervisory system involved in the control of action generates a dysexecutive syndrome.

Attentional episodes Effective cognition requires the sequential resolution of hierarchically organized subproblems, each one with its own relevant inputs, actions, and outcomes. A step-by-step implementation (i.e., episode) of selective attention supports cognition by admitting into the current task set only relevant information (Duncan, 2013).

Attention Networks Model First proposed by Posner and Petersen (1990) and later revised in 2012 (Petersen and Posner), this model identifies three independent but interactive attentional components of *alerting* (defined as the ability to produce and maintain a state of readiness to respond to incoming stimuli), *orienting* (defined as the ability to select some aspects of the information to prioritize its processing), and *executive control of attention* (the ability to select the most appropriate response among competing responses and suppressing the effect distracting information)

Covert or overt attention The allocation of attention towards a region of the visual field with (overt) or without (covert) eye movements.

Dorsal Attention Network (DAN) A bilateral set of frontal and parietal regions showing concurrent activation increases during goal-driven attention (Corbetta & Shulman, 2002).

Decision making The process of deliberating between two or more options that vary in terms of subjective value or environmental relevance.

Direct electrocortical stimulation (DES) A neurosurgical method for functional brain mapping. A small electrical current applied directly to cortex or white matter, causing temporary interruption to the underlying network. If behaviour remains unchanged following stimulation, the structure being probed is considered to be non-essential and resectable (i.e., its resection is unlikely to result in permanent neurological deficit), but if there is a relevant behavioural change with stimulation, the structure is considered to be an essential network component and therefore non-resectable.

Default Mode Network (DMN) A bilateral set of medial regions (including, but not limited to, the ventrolateral prefrontal cortex, and the posterior cingulate cortex) showing concurrent deactivation during task execution in a variety of fMRI or PET studies (Raichle et al., 2001).

Exogenous Attention Often studied using non-predictive visual cues preceding an upcoming visual target, defines the allocation of attentional resources towards a salient stimulus through an automatic, effortless, process.

Endogenous Attention Often studied using predictive visual cues preceding an upcoming visual target, defines the allocation of attentional resources towards a behaviourally relevant stimulus through a voluntary, effortful, process implemented in a top-down fashion.

Fluid and Crystallised intelligence The two components of general intelligence relate to reflecting ability to solve problems by abstraction (fluid) and the ability to solve problems by using general knowledge and previous experience (Cattell, 1963).

Functional MRI An MRI technique that indirectly measures activation of brain regions based on blood flow (BOLD activity) either during the completion of a task (task-based) or at rest (resting state).

Frequency bands Initially discovered in EEG studies, refer to the rhythmic, repetitive activity of brain waves recorded

by electrodes. More recently studied using also other electrophysiological and neurophysiological methodologies. This rhythmic activity is usually divided into bands, including delta (< 4 Hz), theta (4 to 8 Hz), alpha (8 to 15 Hz), beta (15 to 32 Hz), and gamma (> 32 Hz).

g factor A general factor that summarizes an individual cognitive abilities, often measured using IQ tests such as the Wechsler Adult Intelligence Scale, first proposed by Spearman (1927).

Hub area A brain region (or sub-component of a network) critically positioned within the network architecture of the brain due to their high number of contacts with other brain regions (Power et al., 2013).

iEEG Intracranial electroencephalography, an invasive method for recording electrical activity from small populations of neurons in the cerebral cortex and in subcortical structures, performed in patients with intractable epilepsy to localise their epileptic focus. Includes two types of recordings: (a) Electrocorticography (ECOG), recording of cortical activity using electrodes placed directly on the brain surface; and (b) Stereotactic EEG, recording of cortical and subcortical activity using penetrating depth electrodes inserted into the brain (Mukamel & Fried, 2012; Parvizi & Kastner, 2018).

Inhibition The active process of suppressing irrelevant stimuli, task sets, behavioural responses, and unwanted thoughts from capturing resource-limited attentional function and, consequently, affecting performance (Aron, 2007).

Lateralization A principle of brain organization that allocates the implementation of one cognitive function exclusively (or for the major part) to one cerebral hemisphere (e.g., Heilman & Van Den Abell, 1979).

Mind wandering A mental state whereby the focus of attention is internal, self-generated thought (see Mason et al., 2007 for details).

Multivoxel pattern analysis (MVPA) A statistical technique that applies pattern classification algorithms to Blood Oxygenation Level Dependent (BOLD) activity acquired using functional MRI.

Neglect (hemispatial neglect) Failure to spontaneously orient to one side of space. Most often presents as left neglect following right hemisphere damage (Heilman & Watson, 1977).

Neural oscillations Rhythmic patterns of neural activity in the central nervous system that can occur during activation of small-to-large populations of neurons and appear to support cognitive processes.

Phasic alerting Temporary increase in responsiveness produced by the presentation of a sudden stimulus acting as a warning signal (Aston-Jones & Cohen, 2005).

Task set A stimulus–response relationship that includes formation of a criterion to respond to a target, organization of the program to complete a specific task, and adjustments of the automatic processes.

Tonic alerting The ability to produce and maintain a prolonged state of readiness in the attempt to detect rarely occurring stimuli, often characterized by a monotonic decrease in performance as the required task duration increases. Closely related to the concepts of *energization* and *sustained attention* in other attentional models (Aston-Jones & Cohen, 2005).

VAN A set of fronto-parieto-temporal regions lateralized to the right hemisphere showing concurrent activation increases during bottom-up or exogenous attention (Corbetta & Shulman, 2002).

Voxel-based lesion-symptom mapping (VLSM) Univariate method of neuroimaging analysis comparing patients with or without lesions in any given voxel (or region of interest, for the volume of interest-based version of this method) with respect to differences in behavioural measures (t-statistic is computed per voxel: see Bates et al., 2003 for details).

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